

A GUIDE TO PRODUCT QUALITY STANDARDS ENGINEERING GOODS



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ABBREVIATIONS

<i>ABCB</i>	Australia Building Codes Board
<i>ADV</i>	Animal Driven Vehicles
<i>ANSI</i>	American National Standards Institute
<i>APLAC</i>	Asia Pacific Laboratory Accreditation Cooperation
<i>AS</i>	Australian Standards
<i>ASTM</i>	American Society for Testing and Materials
<i>BICON</i>	Bio Security Import Condition System
<i>BIS</i>	Bureau of Indian Standards
<i>CE</i>	Conformité Européenne
<i>CoC</i>	Certificate of Conformity
<i>CPB</i>	Customs and Border Protection
<i>CPSC</i>	Consumer Products Safety Commission
<i>CPSIA</i>	Consumer Product Safety Improvement Act
<i>CPSA</i>	Consumer Product Safety Act
<i>DOT</i>	Department of Transportation
<i>EC</i>	Council Regulation (Regulation of the European Parliament and Council)
<i>EN</i>	European Standards
<i>EU</i>	European Union
<i>FCM</i>	Food Contact Materials
<i>FDA</i>	Food and Drug Administration
<i>FHSA</i>	Federal Hazardous Substances Act
<i>FSSAI</i>	Food Safety and Standards Authority of India
<i>GCC</i>	Gulf Co-operation Council
<i>GSO</i>	GCC Standardization Organization
<i>GPSD</i>	General Product Safety Directive
<i>IS</i>	Indian Standard
<i>ISI</i>	Indian Standards Institution
<i>ILAC</i>	International Laboratory Accreditation Cooperation
<i>ISO</i>	International Organization for Standardization
<i>MLA</i>	Multilateral Recognition Arrangement
<i>MRA</i>	Mutual Recognition Arrangement
<i>MSME</i>	Micro, Small & Medium Enterprises
<i>NABCB</i>	National Accreditation Board for Certification Bodies
<i>NABL</i>	National Accreditation Board for Testing and Calibration Laboratories
<i>NHTSA</i>	National Highway Traffic Safety Administration
<i>OEM</i>	Original Equipment Manufacturer
<i>RoHS</i>	Restriction of Hazardous Substances Directive
<i>SASO</i>	Saudi Standards, Metrology and Quality Organization
<i>QCI</i>	Quality Council of India
<i>QCO</i>	Quality Control Order

BACKGROUND

Uttar Pradesh, the Indian state with a geographical expanse of 2, 40,928 sq.km and a population of 204.2 million people is home to a great diversity in all facets of life. With diverse community traditions and economic pursuits, Uttar Pradesh is also endowed with a beautiful diversity of crafts and industries spread across small towns and districts known for the uniqueness of these products.

To encourage such indigenous products and crafts, the Uttar Pradesh government has initiated the One District One Product (ODOP) Scheme with a vision to create product-specific traditional industrial hubs across 75 districts of Uttar Pradesh that will also promote traditional industries that are synonymous with the respective districts of the state. Although many of the district-specific industries are more of a commonplace but there are several products that are unique to those regions.

Many of these products are GI-tagged; a geographical indication (GI) is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. This becomes a testimony to the original creativity and exclusiveness of some of the finest products produced in the various districts of Uttar Pradesh.

To help in achieving the objectives of the ODOP initiative, the ODOP Cell, Government of Uttar Pradesh, engaged Quality Council of India (QCI) to develop 'Compendiums of relevant Regulations and

Standards' for the products under this initiative. The compendiums have been developed by the Quality Council of India with assistance of the sectoral and product experts. Field visits and virtual interactions with ODOP manufacturers were also undertaken to assess the level of their understanding on relevant standards and certifications. The ODOP Cell, Government of Uttar Pradesh and the EY team at the ODOP Cell also provided continuous support in the development of these compendiums.

The key objective of developing the compendiums is to provide a ready reference of various regulations & standards for the existing ODOP manufacturers and producers and also for the aspiring entrepreneurs.

This compendium brings relevant regulations and standards pertaining to the products falling under the Engineering Goods category.

Uttar Pradesh having traditionally been known for its agrarian prowess has taken promising steps to foster industrial growth. The industrial landscape in the state is dominated by MSME units especially in the light engineering goods categories. A part of the Delhi-Gurgaon-Noida-Ghaziabad auto-cluster is situated in Western Uttar Pradesh and a number of original equipment manufacturers (OEMs) and auto component suppliers have also setup their base at Noida and Ghaziabad. Noida has also emerged as a hub for electronic goods and electrical equipment.

RIM AND AXLE, SHAMLI

1. INTRODUCTION

Shamli is known for the many MSME units involved in manufacturing of Rims and Axles for animal driven vehicles. These units also export their products to various countries apart from supplying to the local demand.

Axle and Rims are important components of any vehicle, be it animal driven (ADV) or run by another kind of fuels. Axle is a shaft used for transmission of energy from engine to the wheels. They can be rolled, forged, hollow or any shape that may be desired while a rim is the "outer edge of a wheel, holding the tire". In today's context a "wheel" and "rim" are used interchangeably but they are not the same. The rim is actually just the outer lip of a wheel.

The origins of the Rim go back to the first pneumatic tires made for bicycles by John Boyd Dunlop. They were simple tubes in shape secured to the wooden outer concave surfaced circumference of the wheel by glue and air pressure pressing them against it. The surface for receiving the tube was not very secure thus causing the tires to sometimes come off the rims. Another Bicycle manufacturer and inventor, Thomas B. Jeffery, developed an improved tire with a wire that was embedded in the rubber of the tire and the wire could be tightened onto the rim. Patented in 1882, it became the ancestor of all clincher tires which are the designs found on modern

bikes and cars. Contemporary clincher tires today have wires embedded on both beads of the tire, so that the wires fit inside the edges of the rim to hold the tire in place when it is fully inflated.

1.1. Rims

The rim is the "outer edge of a wheel, holding the tire". It makes up the outer circular design of the wheel on which the inside edge of the tire is mounted on vehicles such as Animal drawn carts and trolleys, cars, trucks and other heavy machinery.

The rim is an essential part of the wheel because the proper fitting between the tire and wheel wouldn't be possible without it. If the rim size is not compatible with that of the tire, it compromises the performance and stability of the car, leading to a risk of collision. The rims can be classified as below:

a. Safety Rims: Used with tubeless tires, the safety rim helps keep the tire bead (the edge of the tire that sits on the wheel) held to the rim under adverse conditions such as loss of pressure by having a pair of safety humps extending inwardly of the rim toward the other tire bead seat from an outer contoured surface of the rim.

b. Split Rims: Heavier vehicles sometimes use a multi-piece rim assembly. These use a side ring or lock ring to hold the tire in place. The split rims can be seen on tractors, forklifts, and other heavy-duty utility vehicles. Multi-piece wheels can be dangerous if they are suddenly separated. This means they are considered an explosion risk. For this reason, these are usually not encountered on passenger vehicles.

c. Bead Lock Rims: Bead lock rim is a mechanical device that secures the bead of a tire to the wheel of a vehicle. Tires and wheels are designed so that when the tire is inflated, the tire pressure pushes the bead of the tire against the inside of the wheel rim so that the tire stays on the wheel and the two rotate together. In situations where tire pressure is insufficient to hold the bead of the tire in place, a bead lock is needed.

1.2. Axle

An axle is a central shaft for a rotating wheel or gear. On wheeled vehicles, the axle may be fixed to the wheels, rotating with them, or fixed to the vehicle, with the wheels rotating around the axle. In the former case, bearings or bushings are provided at the mounting points

where the axle is supported. In the latter case, a bearing or bushing sits inside a central hole in the wheel to allow the wheel or gear to rotate around the axle. Sometimes, especially on bicycles, the latter type axle is referred to as a spindle. Axles can be divided as:

a. Straight Axle: Single rigid shaft connecting a wheel on the left side of the vehicle to a wheel on the right side. The axis of rotation fixed by the axle is common to both wheels. Such a design can keep the wheel positions steady under heavy stress and can therefore, support heavy loads.

Straight axles are used on trains (i.e., locomotives and railway wagons), for the rear axles of commercial trucks and on heavy duty Animal driven vehicles (ADVs). The axle can optionally be protected and further reinforced by enclosing the length of the axle in a housing.

b. Split Axle: In split-axle designs, the wheel on each side is attached to a separate shaft. Modern passenger cars have split drive axles. In some designs, this allows independent suspension of the left and right wheels and therefore a smoother ride. Even when the suspension is not independent, split axles

permit the use of a differential, allowing the left and right drive wheels to be driven at different speeds as the automobile turns, improving traction and extending tire life.

- c. **Tandem Axle:** Tandem axle is a group of two or more axles situated close together. Truck designs use such a configuration to provide a greater weight capacity than a single axle. Semi-trailers usually have a tandem axle at the rear.
- d. **Transaxle:** Transaxle is a single mechanical device which combines the functions of an automobile's transmission, axle and differential into one integrated assembly. It can be produced in both manual and automatic versions.
- e. **Beam Axle:** Beam axle also known as rigid axle or solid axle is a dependent suspension design in which a set of wheels is connected laterally by a single beam or shaft. Beam axles were once commonly used at the rear wheels of a vehicle, but historically they have also been used as front axles in four-wheel-drive vehicles. In most automobiles, beam axles have been replaced with front and rear independent suspensions.

2. RAW MATERIALS

2.1. The materials used for the manufacture of the rims (along with the wheel) are varied depending on the application. Generally, the main raw materials used are:

- a. Steel plates
- b. Aluminum castings and forming's
- c. Magnesium alloys
- d. For high end applications, even for truck rims, titanium alloys are used

Usually for ADV's, steel is usually used to manufacture the rims

2.2. Raw materials for the axles are:

- a. Carbon steel
- b. Chrome molybdenum steel

3. MANUFACTURING PROCESS

3.1. RIM

- a. **Making a sleeve:** The sheet is cut to a predetermined size and bent and rolled to form a circular sleeve.
- b. **Welding:** After the sleeve is formed, the two free ends are welded together. Extra weld material is removed to get smooth surface.
- c. **Cylindrical Flow spinning:** The sleeve is formed to desired shape/ form by spinning.
- d. **Shaping and Calibration:** After the desired thickness is achieved, the sleeve is shaped to obtain rims on both the sides

with an inner cylindrical wall in the zone of the outer seat and with a radially outer frustoconical wall inclined at an angle corresponding to the standard inclination of the rim seats. The dimensions of the edges are achieved by machining. After this, the rim is calibrated.

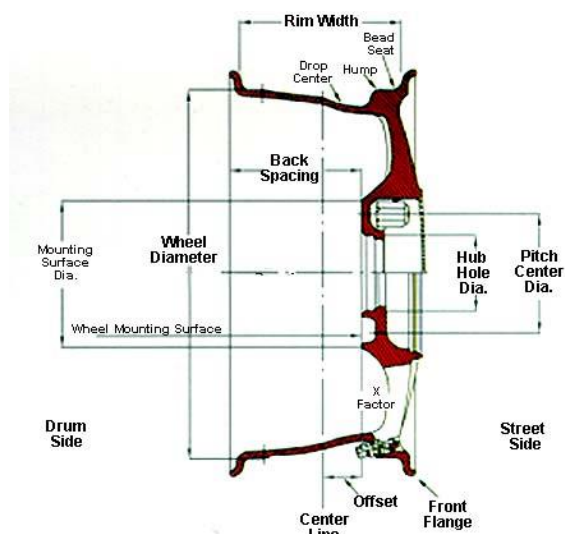


Figure 1: Cross section of a wheel

- e. Fixing to the Wheel:** To support the cylindrical rim structure, a disc is made by stamping a metal plate. It has to have appropriate holes for the center hub and lug nuts. The radial outer surface of the wheel disk has a cylindrical geometry to fit inside the rim. The rim and wheel disk are assembled by fitting together under the outer seat of the rim and then being welded together. The disk is welded in place such that the center of the wheel is equal to the center of

the hub. The distance between the centerline of the rim and the mounting plane of the wheel is called the "offset" and can be positive, negative or zero.

3.2. AXLE

- a. Load calculation:** The axle size depends primarily on the weight it has to support. Usually, these are available from the trailer or automobile manufacturers, or the fabricators who fabricate the trailer.



Figure 2: Axle Forged blanks

- b. Blanks:** Most axles start of as forged blanks. These can be machined to fit any requirement. In a few cases like for Animal Driven Vehicles (ADV) they may be scavenged from automobile parts from old vehicles.



Figure 3: Splines cut on blanks

c. Cutting Splines: Axle splines are designed to mesh with the side spider gears and transfer the torque motion of the differential to the axle. The idea of having a spline on the axle seems contradictory considering the massive amount of torque some of the axles can see. OEM axle splines are rolled into the metal, rather than cut. Rolling is faster and less expensive. This is faster for mass production but allows for more variance. These are cut on lathe machines.

d. Hardening/Heat Treatment: Without this, the axle would twist and therefore, it is an integral part of treatment of axles. This is done by:

- Induction Hardening
- Through Hardening

e. Drilling: The weight from the center of the axle can be removed by drilling.



Figure 4: Drilled out center core

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- The fabricated product should be checked for workmanship
- Check for defects such as dents, deformations, etc.
- There should be no deformation or bending in axles
- Random checking of shapes and dimensions
- Painting quality and uniformity, if any

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organizations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with

several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from

Department of Commerce
[\(https://commerce.gov.in/international-trade/trade-agreements/\)](https://commerce.gov.in/international-trade/trade-agreements/).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	NA	NA	NA

5.2. International

S. No.	Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Regulation (EU) 2021/646	a. European Commission	a. Directives and regulations on motor vehicles, their trailers, systems and components
2	United States of America	a. US Code 127 Vehicle Regulations	a. National Highway Traffic Safety Administration, (NHTSA) b. Department of Transportation (DOT)	a. Vehicle weight limitations-interstate system b. Rim information, marking and rim listing to be etched on the rims.
3	Australia	a. Australian Design Rules, Motor Vehicle Standards Act ,1989 b. Road Traffic (Vehicles) Regulations 2014	a. Department of infrastructure, transport, regional development and communications	a. Motor vehicle safety standards b. Trailers ((VSB1) technical requirements
4	Middle East	a. GSO 02/01/CD/Update for No. 42:2003 b. SASO Certificate of Conformity	a. Saudi Standards, Metrology and Quality Organization (SASO)	a. Vehicle safety guidelines

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
General (Refer Table 1)	IS 12535: 1991: Part 3 (Reaffirmed Year: 2021)	Automotive vehicles - transmission systems - glossary - Part 3 Drive axles definition	Australia	ISO 14400: 2005	Road vehicles - Wheels and rims - Use, general maintenance and safety requirements and out-of-service conditions

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
	IS 4930: 1985 (Reaffirmed Year: 2019)	Axle assembly for pneumatic-wheeled animal-drawn vehicles	-	-	-
Raw Materials (Refer Table 2)	-	-	Middle East	GSO ISO 17900: 2013	Agricultural trailers - Balanced and semi-Mounted trailers - determination of payload, vertical static load
Manufacturing/ Processing (Refer Table 3a. & 3.b)	IS 1628: 1986 (Reaffirmed Year: 2018)	Axle oil	European Union	BS ISO 3894: 2015	Road vehicles. Wheels/rims for commercial vehicles. test methods
	IS 11269: 1985 (Reaffirmed Year: 2019)	Axle assembly for conventional animal-drawn Vehicles	USA	ASTM E2467-05 (2018)	Standard practice for developing axle count adjustment factors
	IS 9436: 2018	Performance requirements and methods of test for wheels of passenger cars, quadri-cycles and mini goods carriers	Australia	AS/NZS 4737: 2002	Motor vehicles - 'I' beam steer axle camber adjustment
	IS 10694: Part 1: 2009 (Reaffirmed Year: 2019)	Automotive vehicles rims- General requirements -	Middle East	GSO ISO 7867-1: 2008	Tires and rims (metric Series) for agricultural tractors and machines – Part 1: Tire designation
Finishing (Refer Table 4)	-	-	USA	ASTM F2803 -09(2015)	Standard test method for evaluating rim slip performance of tires and wheels
	-	-	Middle East	GSO ISO 10392: 2013	Road vehicles - Determination of center of gravity

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	IS 4930: 1985 (Reaffirmed Year: 2019) - Specification for axle assembly for pneumatic-wheeled animal-drawn vehicles	- Scope - Nomenclature
	IS 12535: 1991: Part 3 (Reaffirmed Year: 2021) - Drive Axels definitions	- Glossary of terms common designs of axles for automotive drives - Diagrams

Stage	Code	Scope
	ISO 14400: 2005 - Road vehicles - Wheels and rims - Use, general maintenance and safety requirements and out-of-service conditions	- Requirements for use, general maintenance - Exclusions

6.2. Table 2: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Materials	GSO ISO 17900 – 2013 - Agricultural trailers — balanced and semi-mounted trailers — Determination of payload, vertical static load and axle load	Methods of determination of payloads

6.3. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	IS 10694: Part 1: 2009 (Reaffirmed Year: 2019) - Automotive vehicles rims- General requirements	- Nomenclature and designation - Marking - Referenced documents
	IS 1628: 1986 (Reaffirmed Year: 2018) - Specification for Axle Oil	- Requirements and methods of Sampling and testing - Grades
	IS 11269: 1985 (Reaffirmed Year: 2019) - Specification for axle assembly for conventional animal drawn vehicles	- Scope - Nomenclature and materials used for manufacture - Dimensions

6.4. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	BS ISO 3894: 2015 - Road vehicles. Wheels/rims for commercial vehicles. Test methods	- Wheels/rims for commercial vehicles. Test methods is classified in these ICS categories: 43.040.50 Transmissions, suspensions
	ASTM E2467-05 (2018) - Standard practice for developing axle count adjustment factors	Development of axle count adjustment factors, using axle counts and vehicle classification counts
	AS/NZS 4737: 2002 - Motor vehicles - 'I' beam steer axle camber adjustment	Motor vehicles - 'I' beam steer axle camber adjustment
	GSO ISO 7867-1: 2008 - Tires and Rims (Metric Series) for agricultural tractors and machines – Part 1: Tire Designation, dimensions and marking, and tire/rim co-ordination	- The size designation, the dimensional calculation and the markings of the metric series of tires primarily intended for use on agricultural tractors and machines. - Tire and rim co-ordination

6.5. Table 4: (International Standards for Finishing)

Stage	Code	Scope
Finishing	ASTM F2803-09(2015) - Standard test method for evaluating rim slip performance of tires and wheels	Comparative testing, for driven wheel rim slip, between candidate tire group(s) and a control tire group
	GSO ISO 10392: 2013 - Road Vehicles - Determination of center of gravity	Determining the location of the center of gravity (CG) of a road vehicle

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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1. Product information: <http://odopup.in/en/article/shamli>
2. Product information: [https://en.wikipedia.org/wiki/Rim_\(wheel\)](https://en.wikipedia.org/wiki/Rim_(wheel))
3. Product information: <https://www.cjponyparts.com/resources/types-of-rims>
4. Product information: <http://www.ijsrd.com/articles/IJSRDV3I31467.pdf>
5. Product information: <https://itstillruns.com/tell-its-fake-forgiato-8512720.html>
6. Product information: <https://en.wikipedia.org/wiki/Axle>
7. Product information: <https://en.wikipedia.org/wiki/Transaxle>
8. Product information: https://en.wikipedia.org/wiki/Beam_axle
9. Product information: <https://www.ford-trucks.com/articles/axle-tech-axle-splines/>
10. Product information: <https://www.fordmuscle.com/tech-stories/drivetrain/axles-101-understanding-the-terminology-behind-streetstrip-axles/>
11. Product information: https://hmr.araiindia.com/Control/AIS/PUB_2~17~2012~2~23~46~PM~AIS112_F.pdf
12. Domestic Legislation/Regulation: <http://www.kline.co.in/pdf/weight-restriction.pdf>
<https://www.certification-india.com/en/india-mandatory-bis-certification-for-vehicle-rims-planned/>
13. International Legislation/Regulation:
 - a. EU: https://ec.europa.eu/growth/sectors/automotive/legislation/motor-vehicles-trailers_en
<https://www.legislation.gov.uk/ukxi/2012/2619/contents/made>
 - b. USA: <https://www.law.cornell.edu/uscode/text/23/127>
<https://itstillruns.com/dot-specifications-rims-7423898.html>
 - c. AUS: [https://www.parliament.wa.gov.au/publications/tailedpapers.nsf/displaypaper/3912598a5d3063e216354de648257df100079558/\\$file/2598.pdf](https://www.parliament.wa.gov.au/publications/tailedpapers.nsf/displaypaper/3912598a5d3063e216354de648257df100079558/$file/2598.pdf)
https://www.infrastructure.gov.au/vehicles/vehicle_regulation/bulletin/vsb1/vsb_01_b.asp
[x](#)
 - d. GULF (GCC): https://members.wto.org/cnattachments/2014/tbt/AE/14_4732_00_e.pdf
14. Domestic Standards for Axle and Rims and raw materials: <https://standardsbis.bsbedge.com/>
15. ISO Standards: <https://www.iso.org/obp/ui#search>
16. International Standards:
 - a. EU: <https://www.en-standard.eu/>
 - b. USA: <https://www.en-standard.eu/astm-standards/> , <https://www.astm.org/>
 - c. AUS: <https://www.standards.org.au/>
 - d. GULF: <https://www.gso.org.sa/store/>
17. NABL accredited laboratories: <https://nabl-india.org/>, <https://parakh.ncog.gov.in>

LOCKS & HARDWARE, ALIGARH

1. INTRODUCTION

Locks and assorted hardware of Aligarh are renowned throughout India. Aligarh has been a hub of manufacturing since antiquity and Aligarh's lock making and hardware industry continues to make waves across Indian and global markets. The history of lock industry in India can trace its roots to Aligarh. Lock making was the prime industry in Aligarh and earlier all the locks were handmade locks. Keeping up with the times, this traditional industry embraced modern manufacturing processes and diversified into manufacturing of hardware such as drawer channels, telescopic channels etc.



Due to the easy availability of the raw materials and power supply, Aligarh has emerged as a major business center. Aligarh locks are exported across the world. Today, the city has thousands of manufacturers, exporters and suppliers involved in the brass, bronze, iron and aluminum industries. The different processes of lock-making are carried out in different units.

2. PADLOCK (TUMBLER TYPE LOCK)

2.1. Raw Materials

With the onset of the Industrial Revolution in the late 18th century and the concomitant development of precision engineering and component standardization, locks and keys of increasing complexity and sophistication were manufactured. There are various types of locks available today based on the physical key system such as warded lock, pin tumbler lock, wafer tumbler lock etc. With advances in technology, electronic keys are also being used to provide security solutions in the form of magnetic door locks and incorporating a host of biometric features, such as finger prints to provide cutting edge locks.

The most commonly used however are the tumbler type of locks. A standard five-tumbler key locks are made of various metals such as:

- a. The internal mechanisms of locks are generally made of brass or die-cast zinc.
- b. The cam, which is the tongue that protrudes from the lock to secure it, is usually made of steel or stainless steel.
- c. The outer casing of a lock may be made of brass, chrome, steel,

nickel or any other durable metal or alloy.

2.2. Manufacturing Process

a. Design: Locks come in grades, from low-security to high-security. A low-security lock is generally made from cheaper materials and its parts can be mass-produced. A company that manufactures low-security locks may have two or three available models. Beyond low-security, the lock manufacturer is, generally what is called in industry parlance, an original equipment manufacturer (OEM), meaning that they make the parts for their locks as well as the final products. The process begins with the manufacturer assessing the customer's desired specifications. The customer may order a lock to fit a certain size door, for example and may require that the locks can be opened with a master key.

The lock manufacturer then comes up with the best design for that customer's needs. Sometimes, the lock manufacturers can reverse engineer a lock and cleverly design around the reverse engineered lock in order to not infringe on the design patent, this

happens usually in the case of high security locks.

b. The Key: For a standard five-tumbler key lock, the key is made first. The lock manufacturer buys key blanks and cuts the ridges or combinations in each key. These keys have five bumps on it that are cut to different levels that are designated by numbers. A low cut is one, next up is two, then three. In many cases, there are only four levels, though some manufacturers may use as many as seven. A five-tumbler key lock with four levels in the key, yields approximately 1024 different possible combinations of ridges in the key.

The ridges are listed by the height of each level, yielding what is called the 'combination' for the key. A key with the combination "12341" is cut with the first ridge at level one, the second at two, the next at three, and so on. The lock manufacturer chooses the combinations at random and cuts each key differently.

c. Internal Mechanisms: The internal mechanisms are made next. These have been designed to fit the particular lock. Because the tiny parts, specifically the pins, they must be manufactured to exceedingly fine tolerances.

The machinists may produce a prototype before commencing with the final mass production. Then the machines may be re-calibrated, if necessary. The machining of small brass parts involves many steps. They may be cast, then grooved, ridged, jeweled, and polished. Precision tools handle these jobs, cutting the metal within tolerances of plus or minus 0.001 of an inch.

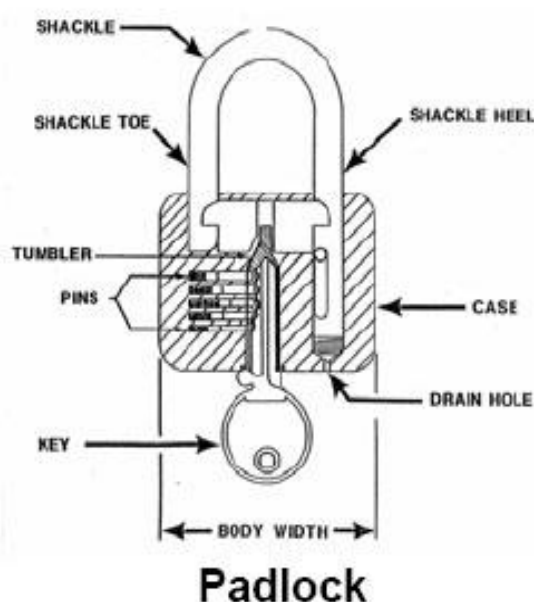


Figure 1: Sectional View 5 - tumbler lock

- d. Other Parts:** The manufacturer also makes the other parts of the lock. The cylinder or plug that the key fits into, guard plates, washers, the bolt or cam, and the casing, are all made according to design specifications, by die-casting and then further machining. The number of parts varies with the design of the lock

but even a small and relatively simple lock may have thirty separate parts, and some of these parts require multiple tooling. The process of making the lock components can take several weeks.

- e. Assembly:** When all the parts are ready, the locks are assembled by hand. Lock workers sit at well-lit tables with a kit of the pieces of the lock in a bin and the key on a stand in front of them. An experienced worker can discern the combination of the key by a mere visual inspection. The worker first fills the plug or cylinder of the lock with the pins that correspond to the combination of the key. The worker then inserts a tiny spring and then the lock pin, using a small tool called an 'assembly pick' to hold the small parts in place. The assembly pick has a small screwdriver on one end and a point on the other, and the worker uses it to prod the delicate parts in where they belong. Once the plug is filled according to the key combination, the worker snaps or screws together the other parts around the lock. Though it is skilled work, it takes no special training and these workers are not locksmiths.

- f. **Final Steps:** Once a lock is fully assembled, the worker checks it with the key for functionality. It may pass to a quality control station at this point and then be dusted or polished. Workers package the completed locks and box them for shipment.

2.3. General Quality Checks

Following are a few visual quality parameters/checks:

- a. Check for the specifications and dimensions
- b. Finishes should be smooth with no sharp edges
- c. Locking and unlocking should be smooth
- d. No rusting or any other residues should be present

3. MORTISE AND CYLINDRICAL LOCKS (Doorlocks & Furniture Locks)

These locks can broadly be categorized into Mortise and Cylindrical locks. Based on their applications they are scaled in sizes as per the standard door thicknesses of a flush door or a furniture cabinet.

3.1. Raw Materials

- a. **Mortise locks** are a very popular and old design having stood the test of time. They are strong and reliable. They are also relatively diverse in their construction.
- b. **Cylindrical locks** with levers usually require a spring cage to lift the lever and keep it

horizontal. These spring cages lead to a bigger rose (circular, finished plate against the door) – typically around 3.5 inches in diameter.

The standard Mortise and Cylindrical lock are made of various metals. The internal mechanisms of locks are generally made of brass or die-cast zinc.

3.2. Manufacturing Process

- a. **Mortise lock:** The mortise is an accumulation of many different pieces that must be placed into a pocket (also known as a mortise) in the door. A pre-assembled lock will also fit into a notch in the door but a mortise must be assembled around the pocket. The parts of the lock are:

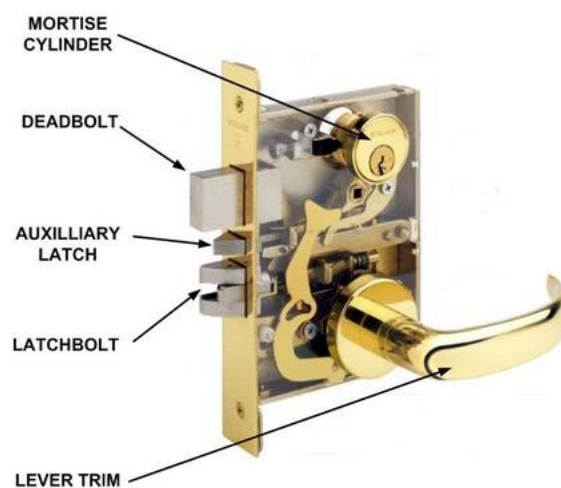


Figure 2: Mortise lock

- **Lock Body:** Comprises of the housing for the deadbolts and the latch bolt
- **Lever Trim/Handle:** The device which has to be

turned in order to retract the latch

- **Through Spindle:** The long rod connecting the lever trim/handle on either side of the door
- **Lock Cylinder:** A threaded cylinder that will secure through the door into the lock body so that when a key is inserted, the door will unlock. A mortise lock cylinder will have a cam, which is a rotating rectangular piece of metal that manipulates the handle's ability to retract the latch.
- **Strike Plate:** This metal fixture is fastened onto the doorway to line up with the latch bolt and potential deadbolt that reside in the lock body of the mortise.

b. Cylindrical lock: A cylindrical lock is designed to be installed through the door with a knob or lever on either side that retract the latch when turned or depressed. The parts of a cylindrical lock are as below:

- **Lock Body:** Also referred as Cylinder, it contains the locking mechanism which is used to unlock the door

- **Handle:** The handles of the cylindrical locks house the lock body
- **Bolt:** The bolt protrudes from the door and engages with the doorframe when the cylinder is in a locked position and recede when the cylinder is unlocked. This is what prevents or allows a door to be opened.
- **Through Spindle:** The long-rod connecting the lever trim/handle on either side of the door.
- **Strike Plate:** This is a metal plate that is fixed to the door frame at the same level as the lock. It contains an opening hole into which the bolt will become engaged when the door is locked. A strike plate sometimes has a lip to help guide the bolt in the right direction.

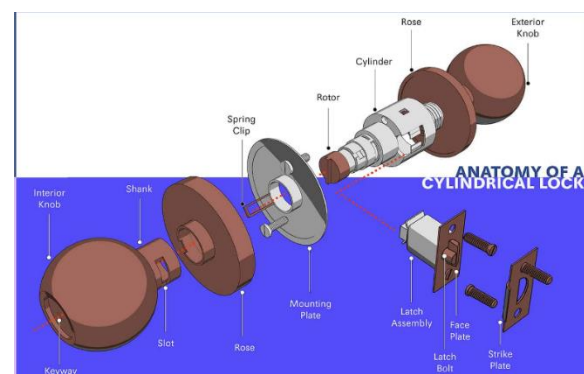


Figure 3: Cut-away section cylindrical lock

3.3. General Quality Checks

- **Mortise lock:** Following are a few visual quality parameters/checks:
 - Check for the specifications and dimensions
 - Finishes should be smooth with no sharp edges
 - Locking and unlocking should be smooth
 - No rusting or any other residues should be present
- **Cylindrical lock:** Following are a few visual quality parameters/checks:
 - Check for the specifications and dimensions
 - Finishes should be smooth with no sharp edges
 - Locking and unlocking should be smooth
 - No rusting or any other residues should be present

4. BICYCLE LOCK

4.1. Raw Material

Bicycle locks, depending on the type, can be manufactured from a wide range of materials such as iron, steel, brass etc.

The most common method for locking bicycles used to be the conventional O-Lock or frame lock. This is a low security mechanism mounted on the frame that immobilizes the rear wheel by moving a steel bolt through the

spokes to prevent motion. An O-lock prevents riding the bicycle but does not, by itself, secure the bicycle to a stationary object. In present times, U shaped locks are more preferred as they enable the bicycle to be tethered to a permanent fixture such as a railing, column etc.

4.2. Manufacturing Process

The steel rods are bent into a characteristic shape and notches are cut at the bottom. These shackles are then heat-treated to increase their strength against attempts at sawing or cutting. Using machinery, the shackles are measured and if need be, corrected to the exact dimensions required.



Figure 4: A typical bicycle lock

The shackles are coated with an anti-rust chemical and then encased in a PVC covering. This is now fitted on to a rectangular block of steel which houses the locking mechanism at the center and has two holes on the edges for the shackles. When the correct key is inserted and turned, an intricate assembly of levers and bolts push

the bolts into the notches cut in the shackles, immobilizing it.

4.3. General Quality Checks

Following are a few visual quality parameters/checks:

- Check for the specifications and dimensions
- Finishes should be smooth with no sharp edges
- Locking and unlocking should be smooth
- No rusting or any other residues should be present

5. DOOR HANDLES

5.1. Raw Materials

Most preferred raw materials for manufacturing of door handles are:

- a. Brass
- b. Stainless Steel

5.2. Manufacturing Process

Though there are several processes used for metal door knobs, including casting where a molten metal is poured into a mold, brass door knobs are typically forged. Forging can be of the following five types namely:

- a. **Open die forging:** Is a hot forming technique used for machining large parts which cannot be formed using the impression forging. This is a faster method but the tradeoff being higher per piece cost and dimensional precision.

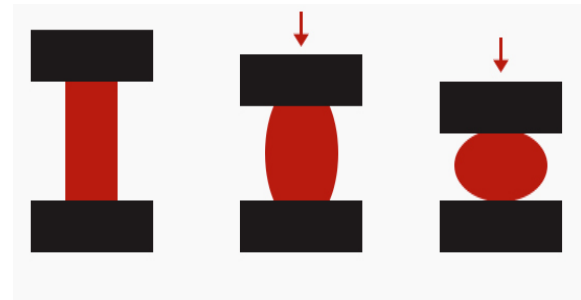


Figure 5: Open die forging

b. Impression die forging:

Utilizes a set of dies with impressions on each. When the dies close, the impressions form a cavity in the shape of the forging. Often, two or more progressive impressions are used, sometimes in conjunction with one or more preforming operations, to form the desired shape. The proper forging temperature improves plastic flow characteristics and reduces the forces on the forging tools.

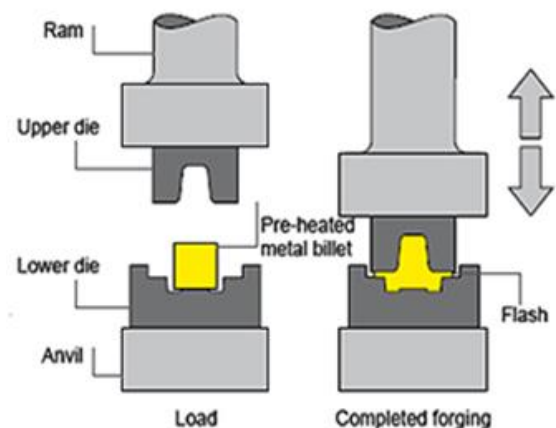


Figure 6: Impression die forging process

- c. **Ring rolling process:** Forms axisymmetric shapes in a hot forming process. The process begins with a "donut" V-shaped

preform which is made by upsetting and piercing operations. The preform is placed over the idler or mandrel roll in a ring rolling mill. The idler roll is moved toward a drive roll which rotates to reduce the wall and increase the diameter while forming.

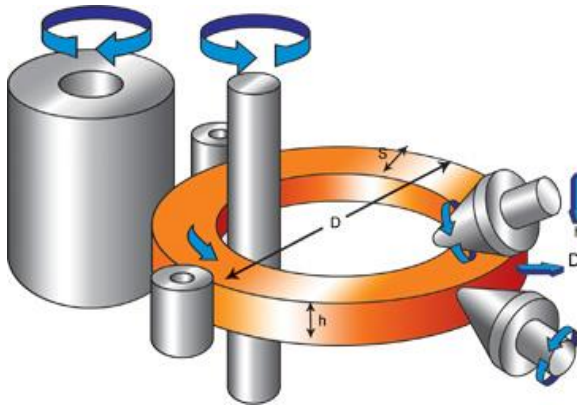


Figure 7: Ring Rolling forging into the desired shape

d. Cold forging: This process employs dies that are sometimes similar to impression dies. The temperature of the

work piece is low enough that scale does not form but the work piece hardens. The lower temperature also promotes greater dimensional accuracy. However, the plastic flow characteristics are not as good at the reduced temperatures, and higher applied forces are required.

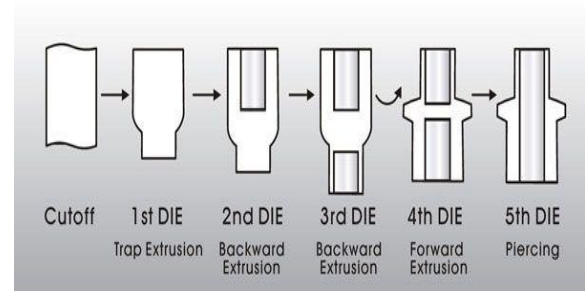


Figure 8: Cold forging process

e. Warm forging: is a modification of the cold forging process where the work piece is heated to a temperature significantly below the typical hot forging temperature.

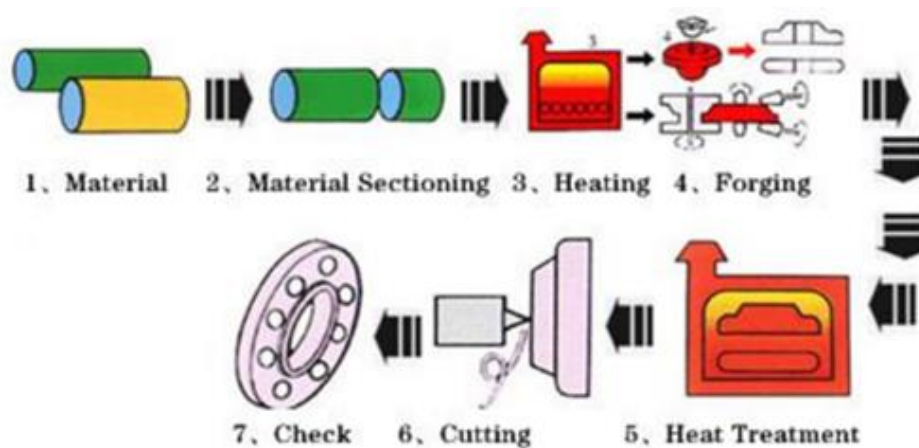


Figure 9: Warm forging process

5.3. General Quality Checks

Following are a few visual quality parameters/checks:

- Check for the specifications and dimensions
- Finishes should be smooth with no sharp edges
- Locking and unlocking should be smooth
- No rusting or any other residues should be present

6. TELESCOPIC CHANNELS

6.1. Raw Materials

Telescopic channels are known by many names such as drawer runner, under mount slide rail, steel ball slide, ball bearing drawer slides, etc. They are widely used in office furniture, kitchen cabinets, home furniture hardware and industrial equipment. The smoothness and loading capacity must be considered when one chooses these furniture fittings. The raw materials for manufacturing these telescopic channels are:

- a. Plastic
- b. Stainless steel
- c. 302 type steel
- d. Galvanized iron

6.2. Manufacturing Process

- a. Major types of drawer slides are 2-section and 3-section which consist of outer, middle, and inner rail. They are widely used in kitchen cabinet, home

furniture hardware, office appliance, and industrial equipment.

b. Telescopic Drawer Slide Channel Production Line Working Flow:

- Usually, the steel to be used arrives at the manufacturing facility in the form of rolls. The sheets are de-coiled and as it unwinds from the parent coil, it is passed through a series of rollers to stretch the material out beyond its yield point (A yield point is the stage at which the solid material changes its shape permanently). This is done to reset any stresses present in the coil during the course of manufacturing.

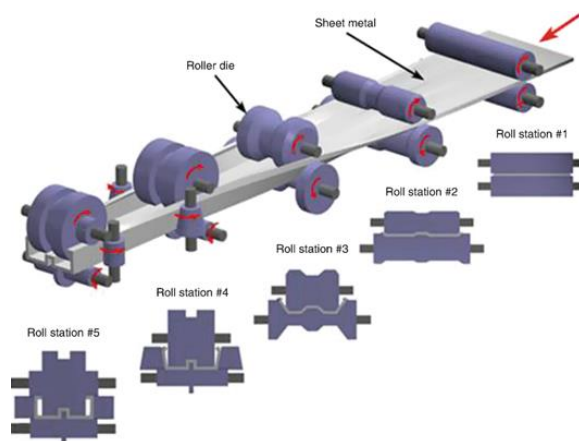


Figure 10: Telescopic drawer production line

- In the second stage, a process known as roll forming is utilized to bend the sheet of steel into a continuous desired section.

The strip passes through sets of rolls mounted on consecutive stands, each set performing only an incremental part of the bend until the desired cross-section (profile) is obtained. Roll forming is ideal for

producing constant-profile parts with long lengths and in large quantities.

- A cutting machine cuts the sections into the desired shapes. These are further given shape by stamping, using various dies.

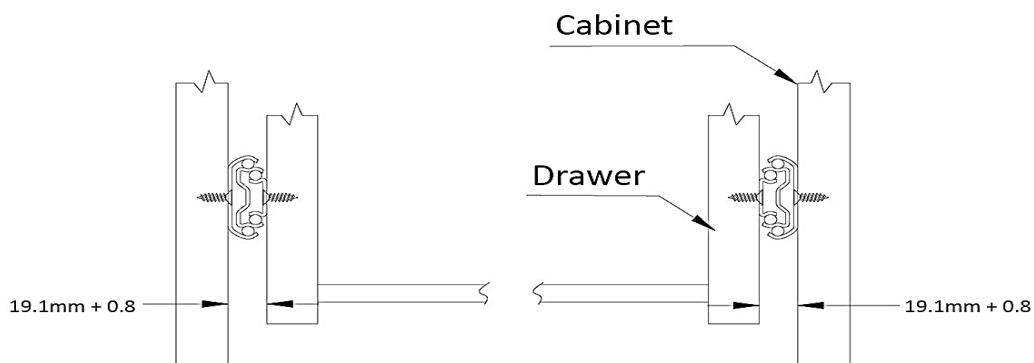


Figure 11: Cross Section view of Channel

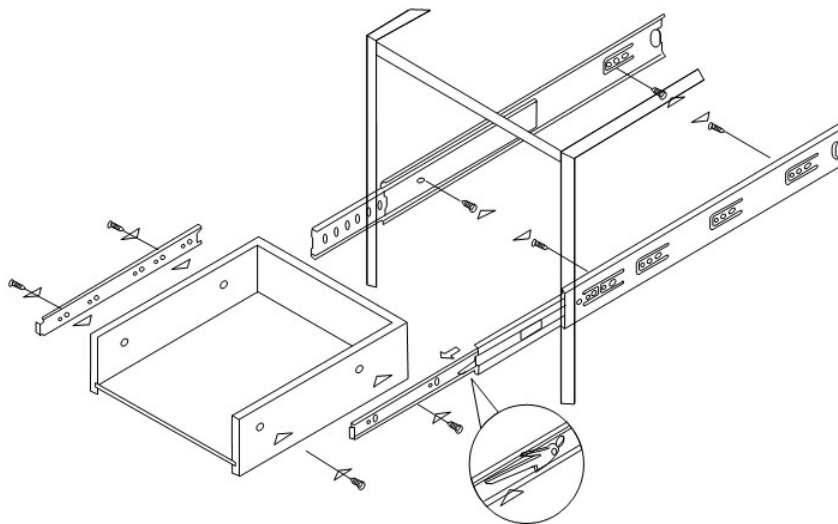


Figure 12: Isometric view of channel operation

6.3. General Quality Checks

Following are a few visual quality parameters/checks:

- Check for scratches and discolouration, if any
- Check for bends or deformations, if any
- Check for smoothness in slide

7. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organizations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (<https://commerce.gov.in/international-trade/trade-agreements/>).

7.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	NA	NA	NA

7.2. International

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Restriction of Hazardous Substances (RoHS) Directive (EC)2015/863 b. Regulation (EU) 2021/854 of 27 May 2021	a. European Commission	a. Compliance on not having more than the prescribed limit for Lead, Chromium and Hexavalent chromium b. Anti-Dumping legislations on imports of stainless steel cold rolled flats originating in India and Indonesia
2	United States of America	a. California Proposition 65, Act of 1986 b. Restriction of Hazardous Substances (RoHS) Regulations	a. California State Government b. Respective State Governments having RoHS regulations in place	a. Any manufacturing unit having more than ten employees are required to label products if they contain carcinogenic products b. Compliance on not having more than the prescribed limit for Lead, Chromium and Hexavalent chromium

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
3	Middle East	a. SASO Certificate of Conformity	a. Saudi Standards, Metrology and Quality Organization (SASO)	a. Product to conform to SASO regulations

8. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
General (Refer table 1)	SP 14: 1976 (Reaffirmed Year: 2018)	Index to IS 1570: Part 1: 1978 (Reaffirmed Year: 2019) steel designations	European Union	DIN 18251,2020	Specifying the dimensions of mortise locks, tubular frame locks and multipoint locks
	IS 7881: Part 1: 1975 (Reaffirmed Year: 2021)	Glossary of terms relating to building hardware: Part 1 locks	Australia	AS 4145.4-2002	Locksets and hardware for doors and windows, Part 2: Mechanical locksets for doors and windows in buildings
	IS 7881: Part 2: 1976 (Reaffirmed Year: 2021)	Glossary of terms relating to building hardware: Part 2 latches	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	IS 2062: 2011 (Reaffirmed Year: 2021)	Hot rolled medium and high tensile structural steel	USA	ASTM A276 - 13a	Standard specification for stainless steel bars and shapes
	IS 4454: Part 1: 2001 (Reaffirmed Year: 2021)	Steel wires for mechanical springs - Part I: Patented and cold drawn steel wires - Unalloyed	USA	ASTM A314 - 19	Standard specification for stainless steel billets and bars for forging
	IS 737: 2008 (Reaffirmed Year: 2018)	Wrought aluminum and aluminum alloy sheet and strip for general engineering purposes	USA	ASTM A555/A555M - 20	Standard specification for general requirements for stainless steel wire and wire rods
	IS 292: 1983 (Reaffirmed Year: 2021)	Leaded brass ingots and castings	USA	ASTM - A484	Standard specification for general requirements for stainless steel bars, billets, and forgings , general delivery requirements
	IS 617: 1994 (Reaffirmed Year: 2020)	Aluminum and aluminum alloy ingots and castings for general engineering purposes	Middle East	GSO ASTM A276: 2009	Standard specification for stainless steel bars and shapes
	-	-	Middle East	GSO ISO 16143-2: 2013	Stainless steels for general purposes - Part 2: Semi-finished

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
					products, bars, rods and sections
Manufacturing/ Processing (Refer Table 3.a & 3.b)	IS 1573: 1986 (Reaffirmed Year: 2021)	Electroplated coatings of zinc on iron and steel	USA	ASTM F3492 - 21	Safety specification for Child Safety Locks and latches for use with cabinet doors and drawers
	IS 16015: 2013 (Reaffirmed Year: 2018)	Mortise Locks with lever mechanism (Vertical type, Sliding door locks and Dead locks)	USA	ASTM A582	Standard specification for free-machining stainless steel bars
	IS 729: 2021	Specification for drawer locks, cupboard locks and box locks	Australia	AS 2837—1986	Wrought alloy steels —stainless steel bars and semi-finished products
	IS 2681: 1993 (Reaffirmed Year: 2018)	Non-ferrous metal sliding door bolts (aldrops) for with padlocks - Specification	Middle East	GSO ISO 1684: 2013	Wire, bar and tube drawing dies - specifications
Finishing (Refer Table 4.a & 4.b)	IS 9844: 1981 (Reaffirmed Year: 2021)	Methods of testing corrosion resistance of electroplated and anodized aluminum coatings by neutral salt spray test	European Union	EN ISO 3651: 2018	Intergranular corrosion testing
	-	-	USA	ASTM - F1577	Standard test methods for detention locks for swinging doors
	-	-	USA	ASTM A800/A800M - 20	Standard practice for estimating Ferrite content of stainless steel castings containing both Ferrite and Austenite
	-	-	USA	ASTM A751 - 20	Chemical analysis of steel products
	-	-	Australia	ISO 15510: 2014	Stainless steels - chemical composition

8.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	DIN 18251,2020 - Locks - Mortise locks and multipoint locks - Terms, definitions and dimensions	This standard is intended as a basis for specifying the dimensions of mortise locks, tubular frame locks and multipoint locks, including their striking plates, as well as rules for dimensioning such locks in order to ensure their compatibility with other components that can be used in combination with the door

Stage	Code	Scope
	SP 14: 1976 (Reaffirmed Year: 2018) - Index to IS 1570: Part 1: 1978 (Reaffirmed Year: 2019) steel designations	- Glossary of terms - Adjunct standards - Referenced documents
	IS 7881: Part 1: 1975 (Reaffirmed Year: 2021) - Glossary of terms relating to building hardware: Part 1 Locks	Glossary of terms
	IS 7881: Part 2: 1976 (Reaffirmed Year: 2021) - Glossary of terms relating to building hardware: Part 2 Latches	Glossary of terms
	AS 4145.4-2002 - Locksets padlocks	Specifies the general design criteria, performance requirements and procedures for testing padlocks for their resistance to physical attacks and corrosion under condition of low to high frequency usage

8.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	IS 2062: 2011 (Reaffirmed Year: 2021) - Hot rolled medium and high tensile structural steel	- Requirements of steel including micro-alloyed steel plates, strips, shapes and sections (angles, tees, beams, channels, etc.), flats, bars, etc., for use in structural work. - Designation of Fe 410 S specified for lock making - Adjunct standards & Referenced documents
	IS 4454: Part 1: 2001 (Reaffirmed Year: 2021) - Steel wires for mechanical springs - Part I: Patented and cold drawn steel wires - Unalloyed	- Steel specification for manufacture of springs for static and dynamic duty - Referenced documents - Test Methodology
	IS 737: 2008 (Reaffirmed Year: 2018) - Wrought aluminum and aluminum alloy sheet and strip for general engineering purposes	- Requirements for general engineering purposes - Referenced documents & adjunct standards - Testing methodology
	IS 292: 1983 (Reaffirmed Year: 2021) - Lead brass ingots and castings	- Specification of grades of lead brass - Referenced documents & adjunct standards - Chemical composition and testing methodology
	IS 617: 1994 (Reaffirmed Year: 2020) - Aluminum and aluminum alloy ingots and castings for general engineering purposes	- Cast aluminum in the form of ingots for general engineering purposes - Referenced documents & adjunct standards

8.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	A276 / A276M – 17 - Standard specification for stainless steel bars and shapes	- Specification for hot-finished or cold-finished bars - Inclusions and exclusions - Referenced documents
	ASTM A314 – 19 - Standard specification for stainless steel billets and bars for forging	Specification stainless steel billets and bars for forging

Stage	Code	Scope
	ASTM A555/A555M – 20 - Standard specification for general Requirements for stainless steel Wire and Wire rods	- General requirements that shall apply to stainless wire and wire rods - Furnishing units
	ASTM - A484 - Austenitic bar and forgings	- General requirements applicable to wrought stainless steel bars, shapes, forgings, and billets or other semi-finished materials - Exclusions - Referenced documents
	GSO ASTM A276: 2009 - Standard specification For stainless steel bars and shapes	- Scope and materials covered - Exclusions
	GSO ISO 16143-2: 2013 - Stainless steels for general Purposes - Part 2: Semi-Finished products, bars, rods and sections	- Technical delivery details - Additional applicability of ISO 404 - Exclusions

8.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	IS 1573: 1986 (Reaffirmed Year: 2021) - Electroplated coatings of zinc on iron and steel	- Requirements for electrodepositing zinc coatings on iron and steel - Exclusions - Terminology specifications - Test methods
	IS 2681: 1993 (Reaffirmed Year: 2018) - Non-ferrous metal sliding door bolts for with padlocks -	- Documents referenced - Finish of Brass sliding doors bolts - Polishing or plating requirements - Specifications for anodizing aluminum
	IS 16015: 2013 (Reaffirmed Year: 2018) - Mortise Locks with lever mechanism (Vertical type, Sliding door locks and Dead locks)	- Significance of ferrite contents - Determination of chemical composition - Determination of ferrite by metallographic means
	IS 729: 2021 - Specification for drawer locks, cupboard locks and box locks	- Dimensions of drawer locks, and their components and tolerances thereon, shall conform to specifications covered in the standard - Dimensions of cupboard locks and box locks having lever mechanism, and their components and tolerances thereon, shall conform to specifications covered in the standard

8.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	ASTM F3492 – 21 - Safety specification for child safety locks and latches for use with cabinet doors and drawers	Defines requirements and test methods for ligature resistant trim on bored locks and mortise locks. These requirements apply to the exposed parts of the lockset on the face of the door in the closed position only
	ASTM A582 - Standard specification for free-machining stainless steel bars	- Specifications for hot-finished or cold-finished bars - Exclusions - Referenced documents
	AS 2837—1986 - wrought alloy steels — Stainless steel bars and semi-finished products	- Specifications for stainless steel bars and other semi-finished products - Referenced documents

Stage	Code	Scope
	GSO ISO 1684: 2013 - Wire, bar and tube drawing dies - Specifications	- Main dimensions for tool types - Codified designation of dies - Referenced documents

8.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 9844: 1981 (Reaffirmed Year: 2021) - Methods of testing corrosion resistance of electroplated and anodized aluminum coatings by neutral salt spray test	- Testing methodology for assessment of quality of coatings - Exclusions - Documents referred and adjunct standards

8.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	EN ISO 3651: 2018 – Intergranular corrosion testing	Determination of resistance to intergranular corrosion of stainless steels ferritic, austenitic and ferritic-austenitic (duplex) stainless steels. Corrosion test in media containing sulfuric acid
	ASTM F1557 -05(2019) - Standard test methods for detention locks for Swinging Doors	- Test methods to evaluate locks against battering devices, prying devices etc. - Exclusions
	A800 / A800M – 20 - Standard practice for estimating Ferrite content of stainless steel castings containing both Ferrite and Austenite	- Test methodology for determination of ferrite content - Exclusions - Referenced documents
	ASTM A751 - 20 - Chemical analysis of steel products	- Test methodology for chemical analysis of steel products - Referenced documents
	ISO 15510: 2014 - Stainless Steels-chemical composition	- Test methodology - Adjunct standards

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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8. Mandatory IS mark: <https://bis.gov.in/index.php/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>
9. Indian Standards under Compulsory Certification: <https://www.bsbedge.com/mandatory-indian-standards>
10. International Legislation/Regulation:
 - a. EU: <https://www.legislation.gov.uk/eur/2006/2023/contents> , <https://www.legislation.gov.uk>
 - b. USA: <https://www.compliancegate.com>, <https://www.accessdata.fda.gov>
11. Domestic Standards for stainless steel rods and raw materials: <https://standardsbis.bsbedge.com/>
12. BIS product conformation overview: <https://bis.gov.in>
13. BIS Certified Labs for Testing products as per IS Standards: http://164.100.105.198:8096/bis_access/
14. Mandatory BIS standards: <https://www.bsbedge.com/mandatory-indian-standards>
15. QCO: <https://steel.gov.in/sites/default/>
16. ISO Standards: <https://www.iso.org/obp>
17. International Standards:
 - a. EU: <https://www.en-standard.eu/>
 - b. USA: <https://www.en-standard.eu/astm-standards/> , <https://www.astm.org/>
 - c. AUS: <https://www.standards.org.au/>
 - d. GULF: <https://www.gso.org.sa/store/>
18. NABL accredited laboratories: <https://nabl-india.org/> , <https://parakh.ncog.gov.in>
19. <https://www.homenish.com/parts-of-a-door-lock/>

CULTIVATOR, GHAZIABAD

1. INTRODUCTION

A cultivator is an agricultural equipment used for secondary tillage. One sense of the name refers to frames with teeth (also called shanks) that pierce the soil as they are dragged through it linearly.



It is an implement for inter cultivation with laterally adjustable tines or discs to work between crop rows. The cultivator stirs the soil and breaks the clods. The tines fitted on the frame of the cultivator, comb the soil deeply creating furrows in the field. A cultivator performs functions intermediate between those of plough and the harrow. However, the primary function of a cultivator is to destroy weeds.

Ghaziabad is known for its industrial prowess and is home to a large number of units involved in fabrication works. The workshops produce a wide range of machinery and associated accessories. Clientele from surrounding states make up for a bulk of the market for these workshops. Cultivators can be of different types:

1.1. Disc cultivator

It is a cultivator fitted with discs.

1.2. Rotary cultivator

It is a cultivator with tines or blades mounted on a power-driven horizontal shaft.

1.3. Tine cultivator

It is a cultivator fitted with tines having shovels.

1.4. Trailed type cultivator

It consists of a main frame which carries a number of cross members to which tines are fitted. At the forward end of the cultivator, there is a hitch arrangement for hitching purpose. A pair of wheels are provided in the cultivator. The height of the hitch is adjusted so that main frame remains horizontal over a range of depth setting. The tines in each row are spaced widely to allow free passage of the soil and trash around them. The tines in subsequent rows are staggered so that the implement can cover the entire width nicely.

1.5. Mounted Cultivator

Tractors fitted with hydraulic lift, operate the mounted type cultivators. A rectangular frame of angle iron is mounted on three-point hydraulic linkage of the tractor. The cross members carry the tines in two staggered lines. Depending upon the type of soil and crop, shovels are chosen for use on the cultivators.

Usually, tractor drawn cultivators are of two types, depending upon the flexibility and rigidity of tines:

- a. Cultivator with spring loaded tines
- b. Cultivator with rigid tines.

1.6. Cultivator with spring loaded tines

A tine hinged to the frame and loaded with a spring so that it swings back when an obstacle is encountered is called spring loaded line. Each of the tine of this type of cultivator is provided with two heavy coil springs, pre-tensioned to ensure minimum movement except when an obstacle is encountered. The springs operate when the points strike roots or large stones by allowing the tines to ride over the obstruction, thus preventing damage. On passing over the obstruction, the tines are automatically reset and work continues without interruption. The tines are made of high carbon steel and are held in proper alignment on the main frame members. This type of cultivator is particularly recommended for soils which are embedded with stones or stumps.

2. RAW MATERIALS

Cultivators are rugged farming implements and they can be made of different materials depending on the soil conditions. The raw materials used can vary from conventional

steels to high carbon steel. Polish and paints are used for providing specific colours.

3. MANUFACTURING

The basic steps in the manufacturing of a cultivator are:

- A simple cultivator is fabricated from iron parts. The basic process commences with the preparation of the frame.
- Rectangular hollow pipe sections are selected as per the length desired and are cut. These are then welded to form a box.
- Next the rigid assembly is fabricated and holes are drilled into it for placing bolts for attaching it to the tractor.
- Steel channel sections are cut and bent to form the cultivator blades, upon which the shoes will be mounted.
- These are placed in an alternating pattern and welded to the main frame.
- The shoes are now installed on the cultivator blades.



Figure 1: Cultivator Frame

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check for the dimensions

- Check the welded joints for proper welding
- The cultivator shoes should be attached firmly to the frame
- The fabricated components should not have any foreign substances adhering to the surface
- Paint should be applied uniformly and should not be flaking off

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country

specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organizations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (<https://commerce.gov.in/international-trade/trade-agreements/>).

5.1. Domestic

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	India	NA	NA	NA

5.2. International

S. No.	Region/Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Safeguard Decision 2013/426 (as amended)	a. European Commission	a. Phytosanitary certification b. Inspection for Animal Diseases
2	United States of America	a. Code of Federal Regulations 9CFR94.1 b. APHIS Regulations	a. APHIS (Animal & Plant Health Inspection Service)	a. Inspection of Animal Diseases b. Phytosanitary requirements
3	Australia	a. Cleanliness requirements b. Bio Security Import Condition System (BICON)	a. Department of Agriculture, Water and Environment	a. Machinery not meeting the acceptable cleanliness standards of the Department of Agriculture and Water Resources may not be permitted entry into Australia b. Biosecurity import conditions

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
General (Refer Table 1)	IS 9818: Part 2: 1981	Glossary of terms related to tillage and inter-cultivation equipment	European union	BS EN 709 - 1997+A4 - 2009	Agricultural and forestry machinery - pedestrian controlled tractors with mounted rotary cultivators, motor hoes, and motor hoe's with drive wheel(s) - safety
	IS 620: 1985 (Reaffirmed Year: 2020)	specification for (reaffirmed 2010) wooden tool handles general requirements	Australia	ISO 4254 - 9: 2018	Agricultural machinery — Safety — Part 9: seed drills
	IS 707: 2011 (Reaffirmed Year: 2021)	Timber technology and utilization of wood, bamboo and cane - Glossary of terms [Adjunct standard to IS 620: 1985 (Reaffirmed Year: 2020)]	-	-	-
	IS 4970: 1973 (Reaffirmed Year: 2020)	Key to identification of commercial timber, [Adjunct to IS 620: 1985 (Reaffirmed Year: 2020)]	-	-	-
	IS 3342: 1998 (Reaffirmed Year: 2019)	Soil working equipment - cultivators, animal drawn (specification of materials for individual components)	-	-	-
	IS 4905: 2015 (Reaffirmed Year: 2020)	Methods of random sampling (Method of selection of tractor-mounted spring-loaded cultivator)	-	-	-
	IS 7201: Part 1: 1987 (Reaffirmed Year: 2019)	Methods of sampling for agricultural machinery and equipment (Hand drawn/animal drawn equipment)	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	IS 1570: Part 1: 1978 (Reaffirmed Year: 2019)	Schedule for wrought steels, steels specified by tensile or yield properties. (Material specification for shovel and sweep of cultivator, C-55 of schedule II)	-	-	-
	IS 2062: 2011 (Reaffirmed Year: 2021)	Hot rolled medium and high tensile structural steel (Material specification, tines)	-	-	-
	IS 2609: 1972	specification for coach bolts (Material	-	-	-

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
	(Reaffirmed Year: 2019)	specification for attaching bolts, Table II)			
Manufacturing/ Processing (Refer Table 3.a & 3.b)	IS 6638: 1972 (Reaffirmed Year: 2021)	Specification for tractor-mounted spring-loaded cultivator	USA	ASAE S225.2	Chisel plows, field cultivators, row crop cultivators, ripper shanks and ground tool mountings
	IS 7565: Part 1: 1975 (Reaffirmed Year: 2019)	Tines for tractor-operated cultivators, Part I: Rigid Tines	Australia	ISO 8945: 1989	Equipment for working the soil - rotary cultivator blades - Fixing dimensions
	IS 7565: Part 2: 1988 (Reaffirmed Year: 2019)	Tines for tractor operated cultivators, Part 2: S-type Tines	Australia	ISO 5680: 1979	Equipment for working soil - Tines and shovels for cultivators - Main fixing dimensions
	IS 4468: Part 1: 1997 (Reaffirmed Year: 2022)	Dimensions for three-point linkage of agricultural wheeled tractors		ISO 730: 2009	Agricultural wheeled tractors — Rear-mounted three-point linkage — Categories 1N, 1, 2N, 2, 3N, 3, 4N and 4
	IS 822: 1970 (Reaffirmed Year: 2019)	Code of procedure for inspection of welds	Middle East	GSO ISO 8945: 2013	Equipment For Working The Soil - Rotary Cultivator Blades - Fixing Dimensions
Finishing (Refer Table 4)	IS 1500: Part 2: 2021	Metallic materials -- Brinell hardness test -- Part 2: Verification and calibration of testing machines	-	-	-

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	AS/NZS 2153.1 – 1997 - Tractors and machinery for agriculture and forestry— Technical means for ensuring safety	- This part of ISO 4254 provides guidelines regarding the prevention of accidents arising from the use of tractors and machinery for agriculture and forestry - It also specifies technical means of improving the degree of personal safety of operators and others involved in the course of normal running, maintenance and use, intended to be carried out by the user of the machinery - This part of ISO 4254 gives general guidelines to be met when designing tractors and machinery for agriculture and forestry
	IS 9818: Part 2: 1981- Glossary of terms related to tillage and inter-cultivation equipment	Glossary of terms

Stage	Code	Scope
	IS 620: 1985 (Reaffirmed Year: 2020) - specification for (reaffirmed 2010) wooden tool handles general requirements	- Specification of types of woods to be used - General requirements of different type of tool handles - Classification of tool handles
	IS 707: 2011 (Reaffirmed Year: 2021) -Timber Technology and Utilization of Wood, Bamboo and Cane - Glossary of Terms	Adjunct to IS 620: 1985 (Reaffirmed Year: 2020), Timber Technology and Utilization of Wood, Bamboo and Cane - Glossary of Terms
	IS 4970: 1973 (Reaffirmed Year: 2020) - Key to identification of Commercial timber	Key to identification of Commercial timber, Adjunct to IS 620: 1985 (Reaffirmed Year: 2020)
	IS 3342: 1998 (Reaffirmed Year: 2019) - Soil working equipment - cultivators, Animal drawn	- Specification of raw materials for various components like seeding attachment, tines etc. - Classification of types of tines
	IS 4905: 2015 (Reaffirmed Year: 2020) - Methods of Random Sampling	Method of selection of tractor-mounted spring-loaded cultivator
	IS 7201: Part 1: 1987 (Reaffirmed Year: 2019) - Methods of sampling for Agricultural machinery and equipment	Methods of sampling for agricultural machinery and equipment (Hand drawn/Animal Drawn equipment)
	BS EN 709 - 1997+A4 – 2009 - Safety requirements, test methods, methods for elimination or reduction of risks arising from their use	Safety requirements, test methods, methods for elimination or reduction of risks arising from their use
	ISO 4254-9 – 2018 - Agricultural machinery — Safety — Part 9: Seed drills	Agricultural machinery — Safety — Part 9: Seed drills

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	IS 1570: Part 1: 1978 (Reaffirmed Year: 2019) - Schedule for wrought steels, steels specified by tensile or yield properties. (Material specification for shovel and sweep of cultivator, C55 of schedule II)	Carbon steel for agricultural implements
	IS 2062: 2011 (Reaffirmed Year: 2021) - Hot rolled medium and high tensile structural steel	- Requirements of steel including micro-alloyed steel plates, strips, shapes and sections (angles, tees, beams, channels, etc.), flats, bars, etc., for use in structural work - steel grades
	IS 2609: 1972 (Reaffirmed Year: 2019) - Specification for coach bolts	Requirements for coach bolts in the size range 6 to 24 mm

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	BS EN 709 - 1997+A4 – 2009 - Agricultural and forestry machinery - Pedestrian controlled tractors with mounted rotary cultivators, motor hoes, motor hoes with drive wheel(s) - Safety	- This European standard specifies safety requirements and testing for design and construction of, pedestrian controlled tractors with mounted rotary cultivators with the cultivator rotating axis horizontal and perpendicular to the direction of motion of the machine, motor hoes and motor hoes with drive wheel(s), all as used in agriculture, forestry, landscaping and gardening (including amenity use) - This European standard does not cover the requirements to fulfil national road traffic regulations such as Lights, steering and braking

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	IS 6638: 1972 (Reaffirmed Year: 2021) - Specification for tractor-mounted spring-loaded cultivator	This standard specifies material, dimensions and other requirements for tractor-mounted spring-loaded cultivators
	IS 7565: Part 1: 1975 (Reaffirmed Year: 2019) - Tines for Tractor-operated cultivators, Part I: Rigid Tines	Specifies materials, dimensions and other requirements for rigid tines for tractor operated cultivators
	IS 7565: Part 2: 1988 (Reaffirmed Year: 2019) - Tines for Tractor operated Cultivators, Part 2: S-type Tines	Specifies materials, dimensions and other requirements for s-type tines for tractor operated cultivators. Three categories, namely, Category 1, Category 2, and Category 3 are specified in this standard
	IS 4468: Part 1: 1997 (Reaffirmed Year: 2022) - Dimensions for three-point linkage of agricultural wheeled tractors	- Specifies the dimensions and requirements of the three-point linkage for the attachment of implements of equipment to the rear of agricultural wheeled tractors - Adjunct Standards
	IS 3342: 1998 (Reaffirmed Year: 2019) - Soil working equipment - cultivators, animal drawn	This standard prescribes material, dimensions and other requirements of animal drawn cultivators

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	ASAE S225.2 - Chisel plows, field cultivators, row crop cultivators, ripper shanks and ground tool mountings	- The purpose of this standard is to provide greater interchangeability of cultivator sweeps, shovels, and points used on various types of foot pieces for row crop cultivators, field cultivators, chisel plows and rippers; to provide a good fit and thereby better mechanical support for sweeps, shovels and points on such foot pieces; and to provide ground clearance between the lowest portion of foot pieces or attaching bolts and the cutting edge of sweeps to accommodate wear - This standard includes the following specifications for row crop cultivators, field cultivators, chisel plows and rippers
	GSO ISO 8945: 2013 - Equipment for working the soil - rotary cultivator Blades - fixing dimensions	This International standard specifies the dimensions for four types of rotary cultivator blades It applies to agricultural rotary cultivator blades with two fixing holes

Stage	Code	Scope
	ISO 8945: 1989 - Equipment for working the soil - rotary cultivator blades - fixing Dimensions	This International Standard specifies the dimensions for four types of rotary cultivator blades. It applies to agricultural rotary cultivator blades with two fixing holes.
	ISO 5680 - Equipment for working soil - Tines and shovels for cultivators - Main fixing dimensions	Main fixing dimensions for interchangeable shovels.

6.6. Table 4: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 1500: Part 2: 2021 - Brinell test for testing hardness of shovel and sweep	Brinell test for hardness of shovel and sweep.

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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3. Mandatory IS mark: <https://bis.gov.in/index.php/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>
4. Indian Standards under Compulsory Certification: <https://www.bsbedge.com/mandatory-indian-standards>
5. International Legislation/Regulation:
AUS: <https://bicon.agriculture.gov.au/BiconWeb4.0>
6. Domestic Standards for Cultivator and raw materials: <https://standardsbis.bsbedge.com/>
7. BIS product conformation overview: <https://bis.gov.in>
8. BIS Certified Labs for Testing products as per IS Standards:
http://164.100.105.198:8096/bis_access/
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12. International Standards:
 - a. EU: <https://www.en-standard.eu/>
 - b. USA: <https://www.en-standard.eu/astm-standards/>, <https://www.astm.org/>
 - c. AUS: <https://www.standards.org.au/>
 - d. GULF: <https://www.gso.org.sa/store/>
13. NABL accredited laboratories: <https://nabl-india.org/>, <https://parakh.ncog.gov.in>

GRILLS, GATES, DOOR & WINDOW FRAMES, SULTANPUR, SHAMLI, FATEHPUR

1. INTRODUCTION

Grillwork is a decorative grating of metal, wood, stone, other material used as a screen, divider, and barrier or as a purely decorative element. With the advances in metalworking, it has led to the prevalence of metal being used increasingly for the manufacture of the doors and window frames as well, instead of previously used wood as the costs for quality wood rose. Metal gates, grilles and other architectural fenestrations in metal works are easily fabricated and last longer than wood which is susceptible to the environment as well as infestation.



Sultanpur is home to a large number of MSME units involved in metal based fabrication works and other engineering products such as electrical machinery,

transport equipment etc. These fabrication units cater extensively to the local market and are known to carefully produce intricate details in metal.

Shamli & Fatehpur are known for the many MSME units involved in fabrication of decorative Grilles and Gates. These units cater primarily to the local markets and are known to produce a wide variety of designs and finishes.

1.1. Gates

Iron front entry doors are often used to improve the appearance of a building or home's exterior. These doors can be fabricated in various styles, decorative patterns and colors which provide for a unique way to customize a home. Aside from adding visual appeal and elegance, these doors offer a number of other benefits like increased security and long-lasting durability. Iron is one of the most resilient and durable materials available for such purposes and are usually more resistant to moisture, rot, fire, and a variety of insects.

1.2. Windows

Windows made of steel and iron fabrication are extremely durable and because of the steel frame and powder coating, they are resistant to decay, weather, and fire. They require minimal upkeep compared to

conventional wooden windows and doors.

1.3. Doors

Metal door frames or iron door frames which are further categorized as pressed steel door frames, standard section window frames, angle iron frames and T-iron frames etc. With the scarcity & high prices of quality wood, metal door and window frames are becoming more widely adopted in the construction industry as it leads to faster production and large numbers can be fabricated and installed in a considerably lesser time than conventional woodwork.

2. RAW MATERIALS

Various raw materials are used for making grilles, gates, windows and doors. The preferred raw materials are:

- Stainless steel
- Wrought iron
- Cast iron
- Brass
- Polish/paint

Wrought iron and Mild steel is the most commonly used material containing between one to four per cent impurities and less than one per cent carbon. It is fibrous and malleable and can be welded. Mild steel is an iron-carbon alloy containing up to about two per cent carbon and has qualities similar to wrought iron.

3. MANUFACTURING PROCESS

3.1. Steel Door

The process of fabrication of a “steel door” is given below:

- The Steel coils are unwound and levelled.
- The steel sheet is fed into a hydraulic embossing press machine and the pattern gets embossed on the steel sheet.
- The embossed pattern is cut and sent to the punching machine where the components are punched using high precision CNC machines to provide for installation of later components such as door hardware.
- In most cases, a press brake bends the vertical edges of the doors along a hardened forming die.
- A separate press with multistage folding tools, bends interlocking edges on the opposite door face so the two pieces can be joined.
- Spot welding is carried out to firm the door with vertical and horizontal edge channel assemblies. If it's a hollow door, a core is placed inside the shell. Further, welding occurs around the perimeter to literally seal the door frame.
- The door is pressed to compact the core inside and the

assembled door is painted in a paint booth.

- The final parts are assembled and stored for further transportation.

3.2. Steel Door & Window Frames

Manufacturing Process of “*steel door and Window Frames*” can be highlighted below:

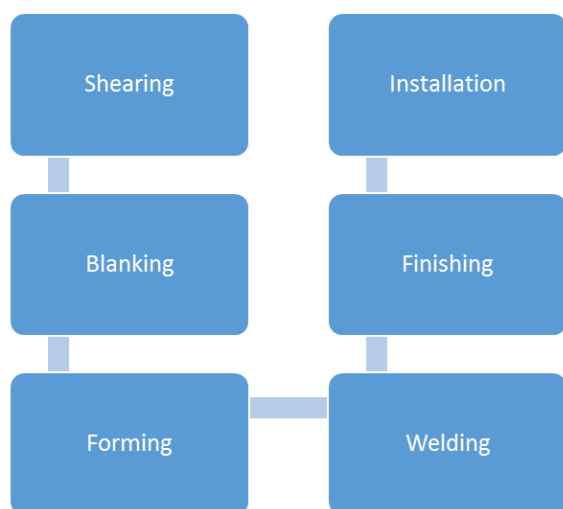


Figure 1: Process Flow Door/Window frame fabrication

3.3. Door & Window Frames

a. Stamped Door/Window Manufacturing Process

- The Steel coils are unwound and levelled.
- The cut steel sections are fed to a high precision CNC punching machine where the components are punched using to provide for installation of later components such as door/window hardware.
- In most cases, a press brake bends the vertical edges of

the doors/windows along a hardened forming die. A separate press with multistage folding tools, bends interlocking edges on the opposite face so the two pieces can be joined.

- The sections now are welded together. The final frame now can be fabricated at the factory or be welded in the field, also referred to as a knocked down frame.
- Post assembly, the welded edges are carefully ground down using machines to achieve a smooth finish. The entire frame is sanded and paint applied.

b. Door/Window Frame Fabrication

- Another method of fabricating is by using metal angle sections for the door/window frames especially for situations where utility is of a more primary concern than aesthetics.
- The MS angle sections are cut in the desired sizes and welded together.
- Holes are drilled along the edge of the door/window frame to receive hinges.

- The door/window frame is cleaned thoroughly and painted.

3.4. Steel Grilles

Manufacturing process of “Grills” can be highlighted below:

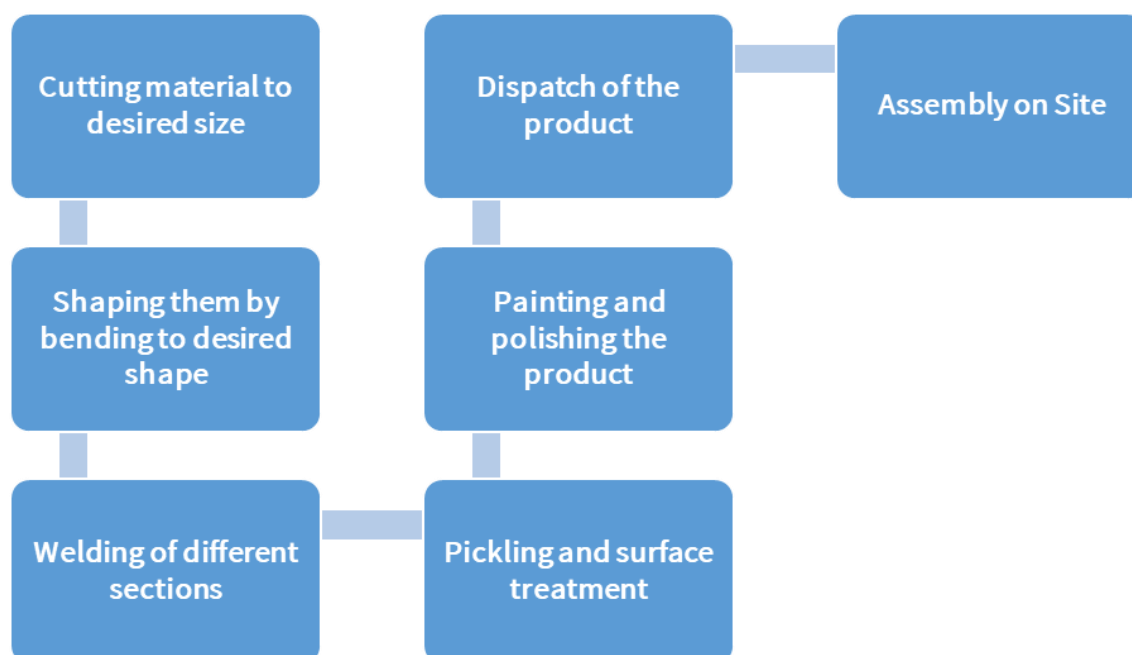


Figure 2: Process Flow Grille

- The frame is fabricated by welding the sections of steel tubes/angles or sections together.
 - The metal segments are chosen as per design and cut and joined together in the desired shape. Though the grilles are primarily used for safety purposes, elaborately designed grilles also double up as works of art rather than being purely a security measure.
- Check the welded joints for proper welding
 - Paint should be applied uniformly and should not be flaking off
 - Check for signs of rust or residues on finished product

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check the dimensions with respect to fenestration sizes

websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organizations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (<https://commerce.gov.in/international-trade/trade-agreements/>).

5.1. Domestic

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	India	NA	NA	NA

5.2. International

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	Europe	a. Commission Implementing Regulation (EU) 2019/1590	a. European Commission	a. (EU) 2016/670 introducing prior Union surveillance of imports of certain iron and steel products originating in certain third countries. b. (EU) 2017/1092 of 20 June 2017 amending Implementing Regulation (EU) 2016/670 c. Système Intégré de Gestion de Licenses (SIGIL) provides quota information on imports of footwear, clothing, steel and wood products.
2	Middle East	a. Saudi Standards, Metrology and Quality Organization regulations. b. SASO Certificate of Conformity	a. Saudi Standards, Metrology and Quality Organization (SASO)	a. Technical Regulation for Doors and Windows b. Structures (excluding prefabricated buildings of heading 9406) and parts of structures (for example, bridges and bridge-sections, lock-gates, towers, lattice masts, roofs, roof structures, doors and windows and their frames and thresholds for doors, shutters, balustrades, pillars) of iron or steel; plates and bars c. Product to conform to SASO regulations.

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
General (Refer Table 1)	IS 10428: 1983 (Reaffirmed Year: 2020)	Glossary of terms relating to doors	USA	ISO 1804: 1972	Glossary of terms relating to doors
	IS 12073: 1987 (Reaffirmed Year: 2018)	Recommendations for modular co- ordination: Co- ordinating sizes for door sets and window sets	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	IS 1161: 2014 (Reaffirmed Year: 2020)	Specification for steel tubes for structural purpose	USA	ASTM E350 - 18	Standard test methods for chemical analysis of carbon steel, low-alloy steel, silicon electrical steel, ingot iron, and wrought iron
	IS 7452: 1990 (Reaffirmed Year: 2017)	Specification for hot rolled steel sections for doors, windows and ventilators	USA	ISO 14284: 1996	steel and iron — Sampling and preparation of samples for the determination of chemical composition
	-	-	USA	ASTM E351 - 18	Standard test methods for chemical analysis of cast iron - All Types
	-	-	USA	ASTM E1806 - 18	Standard practice for sampling steel and iron for determination of chemical composition
Manufacturing/ Processing (Refer Table 3.a & 3.b)	IS 2074: Part 1: 2015 (Reaffirmed Year: 2020)	Specification for ready mixed paint, air drying, red oxide- zinc chrome, priming	European Union	BS 6375-1 -2009	Performance of windows and doors. Classification of weather tightness
	IS 9796: 1981 (Reaffirmed Year: 2018)	Specification for non- weather tight steel doors	European Union	BS 6375-2 -2009	Performance of windows and doors. Classification for operation and strength characteristics and guidance on selection and specification
	IS 1038: 1983 (Reaffirmed Year: 2022)	Specification for steel doors, windows and ventilators	European Union	BS 6510	Specification for steel framed windows and glazed doors
	IS 1948: 1961 (Reaffirmed Year: 2022)	Specification for aluminum doors, windows and ventilators	USA	ANSI/SDI A250.8- 2017	Specifications for Standard steel doors and Frames
	-	-	Australia	AS 2047- 1999	Windows in buildings - Selection and installation
Finishing (Refer Table 4.a & 4.b)	IS 1081: 1960 (Reaffirmed Year: 2022)	Code of practice for fixing and glazing of metal (steel and aluminum) doors, windows and ventilators	Middle East	GSO ISO 3008: 2013	Fire-resistance tests — Door and shutter assemblies

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	IS 10428: 1983 (Reaffirmed Year: 2020) - Glossary of terms relating to doors	Glossary of terms relating to doors
	ISO 1804: 1972 - Glossary of terms relating to doors	Glossary of terms relating to doors
	IS 12073: 1987 (Reaffirmed Year: 2018) - Recommendations for modular co-ordination: Co-ordinating sizes for door sets and window sets	Recommendations for modular co-ordination: Co-ordinating sizes for door sets and window sets

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	IS 1161: 2014 (Reaffirmed Year: 2020) - Specification for steel tubes for structural purpose	This standard covers the requirements for hot finished seamless (HFS), and electric resistance welded (ERW) or high frequency induction welded (HFIW) steel tubes for structural purposes
	IS 7452: 1990 (Reaffirmed Year: 2017) - Specification for hot rolled steel sections for doors, windows and ventilators	This standard lays down the requirements regarding material, nominal dimensions and mass, dimensional and mass tolerances, surface finish and packing for hot rolled steel sections used for doors, windows, ventilators and sashes

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	ASTM E350 – 18 - Standard test methods for chemical analysis of carbon steel, Low-alloy steel, Silicon electrical steel, Ingot iron and wrought iron	Methods cover the chemical analysis of carbon steels, low-alloy steels, silicon electrical steels, ingot iron, and wrought iron
	ISO 14284 – 1996 - Steel and iron — Sampling and preparation of samples for the determination of chemical composition	Gives methods for the determination of the chemical composition of pig iron, cast iron and steel. Methods are specified for use with both liquid and solid metal
	ASTM E351 – 18 - Standard test methods for chemical analysis of cast iron—All Types	Test methods cover the chemical analysis of pig iron, gray cast iron (including alloy and austenitic), white cast iron, malleable cast iron, and ductile (nodular) iron having chemical compositions
	ASTM E1806 – 18 - Standard practice for sampling steel and iron for determination of chemical composition	Covers the sampling of all grades of steel, both cast and wrought, and all types (grades) of cast irons and blast furnace iron for chemical and petrochemical determination of composition

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	IS 9796: 1981 (Reaffirmed Year: 2018) - Specification for non-weather tight steel Doors	- This standard covers the general requirements and basic dimensions of non-weather tight steel doors - This standard does not cover special size doors
	IS 2074: Part 1: 2015 (Reaffirmed Year: 2020) - Specification for ready mixed paint, air drying, red oxide-zinc chrome & priming	- Prescribes requirements methods of sampling and test for the material commercially known as ready mixed paint, air drying, red oxide-zinc chrome priming - The material is used as a primer in the painting system normally followed for enamels for metal surfaces. It is used for the protection of steelwork for household and decorative purposes and also both under marine and inland outdoor conditions
	IS 1038: 1983 (Reaffirmed Year: 2022) - Specification for steel doors, windows and ventilators	- This standard lays down the requirements regarding material, fabrication and finish of steel doors, windows, ventilators and fixed-lights manufactured from rolled steel sections to specified sizes and designs - This standard does not cover steel doors, windows, ventilators and fixed-lights for use in industrial buildings
	IS 12073: 1987 (Reaffirmed Year: 2018) - Recommendations for modular co-ordination: Co-coordinating sizes for door sets and window sets	Specifies the coordinating sizes for door sets and window sets of all materials to be used in buildings and which will fill the coordinating spaces in dimensionally coordinated buildings

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	BS 6375 – 2009 - Performance of windows and doors. (Part 1)	Part 1- Classification of weather tightness and guidance on selection and specification
	BS 6375 – 2009 - Performance of windows and doors. (Part 2)	Part 2 - Classification for operation and strength characteristics and guidance on selection and specification
	BS 6510 - Specification for steel framed windows and glazed doors	- Specifies requirements for the design and fabrication of steel-framed windows and glazed doors - Applicable to windows and doors fabricated into frames in a factory from mild steel profiles, hot-rolled or cold-formed, installed vertically into the external face of buildings, as single or multi-light units
	ANSI/SDI A250.8-2017 - Specifications for standard steel doors and frames	- Specification for standard swinging steel doors and frames offers a variety of choices suitable for any commercial application - Oversizes, design, materials, general construction requirements and finishing of standard steel doors and frames
	AS 2047-1999 - Windows in buildings - Selection and installation	Selection and installation methods

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 1081: 1960 (Reaffirmed Year: 2022) - Code of practice for fixing and glazing of metal (steel and aluminum) doors, windows and ventilators	Covers the recommended methods for fixing and glazing of steel and aluminum doors, windows and ventilators

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	GSO ISO 3008: 2013 - Fire-resistance tests — Door and shutter assemblies	Specifies a method for determining the fire resistance of door and shutter assemblies designed primarily for installation within openings incorporated in vertical separating elements, such as hinged and pivoted doors, horizontally sliding and vertically sliding doors, including articulated sliding doors and sectional doors, steel single-skin folding shutters (un-insulated), other sliding, folding doors, tilting doors, rolling shutter doors and removable panels in walls

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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 - b. American - <https://www.epa.gov/eq/iron-and-steel-manufacturing-effluent-guidelines>
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<https://www.saiglobal.com/pdftemp/previews/osh/as/as2000/2000/2047.pdf>
<https://steeldoor.org/ansi-250-8/>
11. NABL accredited laboratories: <https://nabl-india.org/>, <https://parakh.ncog.gov.in>

ROLLER MILL, GHAZIABAD

1. INTRODUCTION

Rollers and rolling mill (Crushers) are used in sugar industry primarily to extract juice out of sugarcane. The industry in Uttar Pradesh has come up due to the vicinity of sugar mills. India is the second largest producer of sugar in world. However, the size of sugar mills in India are smaller compared to their global competitors.



Roller mill is a form of compression mill that uses a single, double or triple cylindrical heavy wheels mounted horizontally and rotated about their long axis either in opposing pairs or against flat plates, to crush or grind various materials. One of the rollers is run by a motor and the others are rotated by friction as the material is drawn through the gap between the rollers. Roller mills use the process of stress (which is applied by the rotating wheels) and attrition in milling of solids in suspensions, pastes or ointments and some solid materials. The rollers rotate at different speeds and the material is sheared as it passes through the gap.

The rollers for the sugar industry have an additional cutter roller which cuts the cane into segments before it starts crushing the sugarcane. The rollers of the mill are designed to crush sugarcane of different quality (type) of sugarcane. To obtain the desired particles size, the following should be controlled:

- The gap between the rollers
- The speed differential between the rolls
- Feed rate

The design of a roller mill differs depending upon its application and they are usually named according to the work they perform e.g.

- **Crackers:** A mill used to crack grain or other types of friable materials.
- **Flakers:** Mills used to flake grains or other materials
- **Grinders (Rollers mill grinders):** Roller mills that are used to grind in a feed mill.
- **Crimpers:** Roller mills that crimp grains
- **Cracker:** Crackers a roller mills which cracks and at the same time crimp grains.
- **Crumbler:** A roller mill specially designed for breaking up pellets into smaller particles.

These are used in a wide range of industries ranging from metal working, agriculture, food processing, jewelry etc.

Ghaziabad is known for its industrial prowess and is home to a large number of units involved in fabrication works. The units

produce a wide range of machinery and associated accessories. The largest clientele is from the local cane crushing units and goods are also sent to other cane crushing units across the country.

2. RAW MATERIALS

The main raw materials used for the construction of rolling mills are:

a. Rollers

- Steel
- Ceramic

b. Machinery Frame, gears, bearings and housing

- Wrought iron
- Cast iron and
- Steel sheets
- High Speed steel
- Graphite cast steel

c. Polish and Paints

3. MANUFACTURING PROCESS

Stages involved in the making of a Roller mill can be briefly explained as below:

3.1. Designing

Depending on the intended use, the designs are firmed up. The rolling mills can be of the following types:

- Two high rolling mills
- Three high rolling mills
- Four high rolling mills
- Cluster rolling mills
- Planetary rolling mills
- Tandem or Continuous mills

3.2. Frame and Foundations

The foundations are cast and the frames installed. The frames are

erected and fixed to the foundations.

3.3. Rollers

The rollers are an integral part of a Rolling mill and are mounted on the shaft. The Shaft is then mounted on the frame and to it are attached gears for transmission of power using a belt assembly to run the equipment. These rollers can be of various shapes and sizes based on their intended function. The rollers need to withstand high pressures and also be resistant to corrosion.

3.4. Testing and Commissioning

The entire machinery is run and tested to check for any defects and other compliances to requirements laid down during the design phase.



Figure 1: Industrial Use Rollers



Figure 2: A very basic Rolling Mill

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- The fabricated rollers should be checked for workmanship and damages
- Check for rust or any contamination
- Check for accuracy in the assembly
- Check for uniformity of polish/paint

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may

be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organizations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce

(<https://commerce.gov.in/international-trade/trade-agreements/>).

5.1. Domestic

S. No	Country	Regulation	Regulatory Body	Remarks
1	India	a. Food Safety and Standard Act 2006, Rules and Regulations; 2011	a. Food Safety and Standards Authority India (FSSAI)	a. Licensing requirements for food processing industry b. Food Safety compliances by food processing industries

5.2. International

S. No.	Region/Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) regulation (EC) 1907/2006 b. Machinery Directive, EU 2006/42/EC	a. European Commission b. European Chemicals Agency (ECHA)	a. Hazardous chemicals and heavy metals b. CE marking, Traceability, Safety c. GMP for materials coming in contact with food
2	United States of America	a. US Customs and Border Protection (CPB) regulations	a. Customs and Border Protection (CPB)	a. Labelling of product b. Hazardous chemicals and metals

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
		b. California Proposition 65, Act of 1986	b. California State	
3	Australia	a. Biosecurity Act, 2015	a. Department of Agriculture and Water	a. Cleanliness requirements for imported machinery used in field testing/factory testing
4	Middle East	a. Technical Regulations b. SASO Certificate of Conformity	a. Saudi Standards, Metrology and Quality Organization (SASO)	a. Technical regulations for equipment safety b. Product to Conform to SASO Regulations

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
General (Refer Table 1)	IS 16818: 2018	Safety of machinery - Emergency stop function – Principles for design	Australia	AS/NZS 4024.1604 - 2019	Safety of machinery, Part 1604: Design of controls, interlocks and guarding - Emergency stop - Principles for design (ISO 13850:2015 (ED. 3.0) MOD)
	-	-	Middle East	GSO ISO 13850: 2011	Safety Of Machinery - Emergency Stop - Principles For Design
Raw Materials (Refer Table 2.a & 2.b)	IS 1570: Part 2: 1979 (Reaffirmed Year: 2016)	Schedules for wrought steels - Part 2: carbon steels (unalloyed steels)	European Union	BS EN 17449: 2022	Safety of machinery. Safety requirements to finishing lines for metal strip
	IS 3658: 1999 (Reaffirmed Year: 2020)	Code of practice for liquid penetrant flaw detection	USA	ASTM E350 - 18	Standard test methods for chemical analysis of carbon steel, low-alloy steel, silicon electrical steel, ingot iron and wrought iron
	-	-	Middle East	GSO ISO 4987: 2015	steel castings - Liquid penetrant inspection
Manufacturing/ Processing (Refer Table 3.a & 3.b)	IS 11201: 2018	Cane crushing rollers for Sugar Industry-specifications	Middle East	GSO ISO 6743-9: 2008	Lubricants, industrial oils and related products (Class L)
	IS 3612: 2014 (Reaffirmed Year: 2020)	Crushing and grinding equipment classification and methods for measuring capacity	-	-	-

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
Finishing (Refer Table 4.a & 4.b)	IS 7743: 2006 (Reaffirmed Year: 2018)	Recommended practice for magnetic particle testing and inspection of steel forgings	Australia	AS ISO 9934.3 - 2020	Non-destructive testing - Magnetic particle testing, Part 3
	IS 8791: 1978 (Reaffirmed Year: 2020)	Code of practice for ultrasonic flaw detection of ferritic steel forgings	Middle East	GSO ISO 9934-3: 2013	Non-destructive Testing - Magnetic particle testing - Part 3: Equipment

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	IS 16818: 2018 - Safety of Machinery — Emergency stop function — Principles for design	Principles of design for emergency stoppage of machinery
	AS/NZS 4024.1604 – 2019 - Safety of machinery, Part 1604: Design of controls, interlocks and guarding - Emergency stop - Principles for design (ISO 13850:2015 (ED. 3.0) MOD)	- Safety of machinery - Emergency stop
	GSO ISO 13850: 2011 - Safety of machinery - Emergency stop - Principles for design	Functional requirements and design principles for the emergency stop function on machinery, independent of the type of energy used to control the function

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	IS 1570: Part 1: 1978 (Reaffirmed Year: 2019) Part 2 - Schedules for wrought steel, Carbon steels unalloyed	- Specified manganese and carbon limits - Exclusions and tests
	IS 3658: 1999 (Reaffirmed Year: 2020) - Code of practice for liquid penetrant flaw detection test	- Procedures for detection of flaws using liquid penetrant - Does not constitute as a means of acceptance or rejection of product Terminology

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	BS EN 17449:2022 - Safety of machinery. Safety requirements to finishing lines for metal strip	- General safety requirements for finishing lines for metal strip - Significant hazards, hazardous situations and events relevant for finishing lines - Requirements to be met by the manufacturer to ensure the safety of persons and property during transport, commissioning, operation and de-commissioning, as well as in the event of foreseeable failures

	ASTM E350 – 18 - Standard test methods for chemical analysis of carbon steel, low-alloy steel, silicon electrical steel, ingot iron and wrought iron	- Methods for the chemical analysis of metals and alloys - Classification of methods as Referee methods to test for compliances - Laboratory guidelines
	GSO ISO 4987: 2015 - Steel Castings - Liquid penetrant inspection	Liquid Penetration test for forgings

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	IS 11201: 2018 - Cane Crushing Rollers For Sugar Industry	- Physical properties - Shell and shaft material specifications - Terminology - Referenced documents
	IS 3612: 2014 (Reaffirmed Year: 2020) - Crushing And Grinding Equipment Classification And Methods For Measuring Capacity	- Terms, Classification and Nomenclature - Sizes and methods of measuring capacity of crushing and grinding equipment

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	GSO ISO 6743-9: 2008 - Lubricants, industrial oils and related products (Class L) - (Classification – Part 9: Family X (Greases))	Classification of greases as per operating conditions

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 7743: 2006 (Reaffirmed Year: 2018) - Recommended practice for magnetic particle testing and inspection of steel forgings	- Method of magnetic testing for ferromagnetic steel forgings - Terminology - Referenced documents
	IS 8791: 1978 (Reaffirmed Year: 2020) - Code of practice for ultrasonic flaw detection of ferritic steel forging	- Terminology - Equipment characteristics - Estimation of flaw sizes - Test reports

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	AS ISO 9934.3 – 2020 - Non-destructive testing - Magnetic particle testing	Magnetic particle testing
	GSO ISO 9934-3: 2013 - Non-destructive testing - Magnetic particle testing - Part 3: Equipment	Magnetic particle testing

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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STAINLESS STEEL ROD, GHAZIABAD

1. INTRODUCTION

Stainless steel is an alloy containing iron, relatively low concentration of carbon, a minimum 10.5% chromium and in cases up to 30% nickel. However, it usually comprises of 18% chromium and 8% nickel for most applications. Different types of stainless steel include the elements carbon, nitrogen, aluminum, silicon, sulfur, titanium, nickel, copper, selenium, niobium, and molybdenum. Specific types of stainless steel are often designated by their American Steel and Iron Institute (AISI) three-digit number.

Stainless steel rods are manufactured for a wide range of purposes and are favored in the construction industry for the superior finish and aesthetic appeal.



Building a stainless steel product from the ground up involves a lot of work, technology and science. Even though there are billions of users of stainless steel items around the world, there are very few people (relatively) who know what goes into making a stainless steel product from scratch. The manufacturing process of a stainless steel product is fascinating and it gives one an

insight into how far the ability to craft materials has evolved since the early days of the Industrial Revolution.

Ghaziabad is home to a large number of units producing a wide range of specialty long steel, tool steel and various alloy steels for the industry. The units cater to not only domestic but the international markets across industry segments such as automotive, oil & gas and aerospace amongst a few.

2. RAW MATERIALS

For the production of stainless steel rods the basic raw materials are:

- Iron
- Alloying elements such as chromium, nickel, silicon, manganese, nitrogen and carbon.

The properties of the final alloy can be fine-tuned by altering the amounts of the various elements.

3. MANUFACTURING PROCESS

The process of manufacturing stainless steel bars starts with the manufacture of stainless steel itself. The process can be broadly classified into six major steps as below:

3.1. Melting and Casting of Stainless Steel

The Stainless steel manufacturing process consists of two separate channels based on the production of the stainless steel itself:

a. Primary stainless-steel

manufactures: The raw material consists of iron ore, steel scrap and alloying elements like chromium, nickel, manganese and many more depending upon the required finished quality. Large Electric furnaces having capacities up to 50 tons or more are employed for manufacturing Steel. The alloying elements are added in the form of Ferro alloys.

b. Secondary stainless-steel

manufacturers: The raw material in this route is mainly stainless steel scrap. Induction furnaces ranging from 500 kg. to above 50 tons are used for manufacturing. As there is a little possibility of composition control in the Induction Furnace, control on the quality of stainless steel scrap is of great importance to the finished product requirements. Very little amount iron ore and alloying elements in the form Ferro alloys are added.

In both the processes, once the melting is complete, the molten steel is cast (molded) into desired semi - finished forms. Some of the most common forms or shapes include slabs, blooms (rectangular shapes), billets (these could either be round or square), rods and tube rounds.

3.2. Forming

In the second stage, the semi-finished steel shapes undergo a series of forming operations. For instance, the stainless steel is hot rolled (heated and passed through enormous rolls). The blooms and billets mentioned above are converted to bars / rods in the rolling mills to a certain industry defined standard diameters.

The slabs on the other hand are formed into plates, strips or sheets. It is very common to turn semi-finished steel shapes into bars as it is the most versatile stainless-steel form. The reason why it is versatile is because it comes in all grades and sizes. There are different shapes of bars like round, squared, octagonal and hexagonal, each suitable for a different type of application.



Figure 1: Forming Steel

3.3. Heat Treatment

The next step is the heat treatment. As the name clearly suggests, the various stainless steel forms undergo a thorough annealing process during this step. Annealing is another name for heat treatment where the stainless steel is heated and cooled in a controlled environment. The purpose of this heat treatment is to relieve the pent up stress inside the stainless steel and soften the material to make it more suitable for a wide variety of applications. This is a critical step and needs to be monitored very minutely as even the slightest of changes in the temperature, pressure, duration or cooling rate could result in a product which is different from the desired product

3.4. Descaling

During the annealing process, a certain amount of scale appears on the surface of the stainless steel. This scale can be removed using a number of different processes that are collectively known as descaling. Pickling is one of the more common methods of carrying out the descaling process.

3.5. Cutting

The semi-finished, heat treated and descaled stainless steel forms are cut into specific shapes in this step. Mechanical cutting is performed with

the aid of guillotine knives, blanking, nibbling and high speed blades.

3.6. Finishing

Finishing is applied to help the stainless steel product achieve its signature aesthetically appealing appearance. Finishes are also needed to make the stainless steel product smooth and easier to clean, which is a major requirement in sanitary applications.

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Checking for the shape and dimensions
- Check for depressions, deformations, etc.
- Check for surface finish. The surface of finished rods shall be free from cracks, ripples, burrs, pits, laminations or any injurious fabrication, handling or straightening marks

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade

websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organizations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (<https://commerce.gov.in/international-trade/trade-agreements/>).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	a. Quality Control Order SO 4637 (E)	a. BIS	a. Conformation to IS 6603 (stainless steel bars and Flats)

5.2. International

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Restriction of Hazardous Substances (RoHS) Directive (EC) 2015/863 b. Regulation (EU) 2021/854 of 27 May 2021	a. European Commission	a. Compliance on not having more than the prescribed limit for Lead, Chromium and Hexavalent chromium b. Anti-Dumping legislations on imports of stainless steel cold rolled flats originating in India and Indonesia
2	United States of America	a. California Proposition 65, Act of 1986 b. Restriction of Hazardous Substances (RoHS) Regulations	a. California State b. Respective State Governing bodies	a. Any manufacturing unit having more than ten employees are required to label products if they contain carcinogenic products b. Compliance on not having more than the prescribed limit for Lead, Chromium and Hexavalent chromium

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
General (Refer Table 1)	IS 1956: Part 3: 2019	Glossary of Terms Relating to Iron and Steel Part 3 Long Products (Including Bars, Rods, Sections and Wires) (Second Revision)	European Union	EN 10088-2: 1995	Technical delivery conditions for semi-finished products, bars, rods and sections for general purposes

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
	IS 8910: 2022	General technical delivery requirements for steel and steel products	USA	ASTM - A484	Standard specification for general requirements for stainless steel bars, billets, and forgings, general delivery requirements
	SP 17: 1979 (Reaffirmed Year: 2018)	Index to IS 1956, Glossary of terms relating to iron and steel	-	-	-
	IS 1956: Part 1: 1976	Glossary of terms relating to iron and steel Part I General metallurgy, heat treatment and testing	-	-	-
	SP 14: 1976 (Reaffirmed Year: 2018)	Index to steel designations	-	-	-
	IS 1956: Part 3: 2019	Glossary of Terms Relating to Iron and Steel Part 3 Long Products (Including Bars, Rods, Sections and Wires) (Second Revision)	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	IS 6603: 2001 (Reaffirmed Year: 2019)	Stainless steel bars and flats, specifications	USA	ASTM A276 - 13a	Standard specification for stainless steel bars and shapes
	IS 6527: 1995 (Reaffirmed Year: 2019)	Specifications- stainless steel wire rods	USA	ASTM A314 - 19	Standard specification for stainless steel billets and bars for forging
	IS 14652: 1999 (Reaffirmed Year: 2018)	18 percent nickel maraging steel bars and rods — specification	USA	ASTM A555/A555 M - 20	Standard specification for general requirements for stainless steel wire and wire rods.
	-	-	Middle East	GSO ASTM A276: 2009	Standard specification for stainless steel bars and shapes
	-	-	Middle East	GSO ISO 16143-2: 2013	stainless steels for general purposes - Part 2: Semi-finished products, bars, rods and sections

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
Manufacturing/ Processing (Refer Table 3.a & 3.b)	IS 13177: 1991 (Reaffirmed Year: 2021)	Passivation of stainless steel parts - specification	USA	A 276 - 13a	Standard specification for stainless steel bars and shapes
	IS 1570: Part 1: 1978 (Reaffirmed Year: 2019) (Part 5)	Schedules for wrought steels- stainless and heat resistant steels	USA	ASTM A493 - 16	Standard specification for stainless steel wire and wire rods for cold heading and cold forging
	IS 13147: 1991 (Reaffirmed Year: 2021)	Black oxide coating on stainless steel parts - specification	USA	ASTM A582	Standard specification for free-machining stainless steel bars
	-	-	USA	ASTM - A484	Austenitic bar and forgings
	-	-	Australia	AS 2837 - 1986	Wrought alloy steels — stainless steel bars and semi-finished products
	-	-	Middle East	GSO ASTM - A484/A484 M - 2016	Standard specification for general requirements for stainless steel bars, billets and forgings
	-	-	Middle East	GSO ISO 1684: 2013	Wire, bar and tube drawing dies - specifications
Finishing (Refer Table 4.a & 4.b)	IS 3711: 2020	Steel and steel products- Location and preparation of samples and test pieces for mechanical testing	European Union	EN ISO 3651: 2018	Intergranular corrosion testing
	IS 13015: 1991 (Reaffirmed Year: 2018)	Steel products- macro etch testing, specifications for inspection and rating	USA	ASTM A380	Substantial contamination, heat scale or heat tinting through passivation of pickling
	IS/ISO 14284: 1996 (Reaffirmed Year: 2021)	Steel and iron - Sampling and preparation of samples for the determination of chemical composition [MTD 4: wrought steel products]	USA	ASTM A800/A800 M - 20	Standard practice for estimating ferrite content of stainless steel castings containing both ferrite and austenite
	IS 15839: 2021	Determination of ferrite content in austenitic stainless steel castings [MTD 22: metallography and heat treatment]	USA	ASTM A751 - 20	Chemical analysis of steel products

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
	ISO 15510: 2014	Chemical composition of stainless steels	Australia	ISO 15510: 2014	Stainless steels - Chemical composition

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	IS 1956: Part 3: 2019 - Glossary of Terms Relating to Iron and Steel Part 3 Long Products (Including Bars, Rods, Sections and Wires) (Second Revision)	Glossary of Terms Relating to iron and steel, long products (Including bars, rods, sections, wires)
	IS 8910: 2022 - General technical delivery requirements for steel and steel products	- General technical delivery requirements for all steel products covered by ISO 6929 - Exclusions
	SP 17: 1979 (Reaffirmed Year: 2018) - Index to IS 1956, Glossary of terms relating to iron and steel	Glossary of terms, giving clause and part numbers of IS 1956
	IS 1956: Part 1: 1976 - Glossary of terms relating to iron and steel Part I General metallurgy, heat treatment and testing	Glossary of terms used in Metallurgy, Heat Treatment and testing
	SP 14: 1976 (Reaffirmed Year: 2018) : Index to IS 1570: Part 1: 1978 (Reaffirmed Year: 2019) steel Designations	Glossary of terms
	EN 10088-3 1995 - Stainless steels technical delivery conditions for semi-finished products, bars, rods, wire, sections and bright products of corrosion resisting steels for general purposes	- Stainless steels technical delivery conditions for semi-finished products, bars, rods, wire, sections - Bright products of corrosion resisting steels for general purposes technical delivery conditions
	EN 10088-3 - Stainless steels technical delivery conditions for sheet/plate and strip of corrosion resisting steels for general purposes	- Delivery conditions for hot or cold rolled sheet/plate and strip of standard grades and special grades of corrosion resisting stainless steels for general purposes - Exclusions
	ASTM - A484 - Standard specification for general Requirements for stainless steel bars, billets, and forgings	- General requirements that shall apply to wrought stainless steel bars, shapes, forgings, and billets or other semi-finished materials - Furnishing Units

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	IS 6603: 2001 (Reaffirmed Year: 2019) - Stainless steel bars and flats — specification	- Requirements for non-precipitation hardening wrought stainless steels in the form of bars and flats - Terminology - Supply of materials - Manufacturing processes and chemical composition

Stage	Code	Scope
	IS 6527: 1995 (Reaffirmed Year: 2019) - Stainless steel wire rods, specifications	- Requirements for stainless steel wire rods - Exclusions & referenced documents - Chemical compositions and freedom from defects
	IS 14652: 1999 (Reaffirmed Year: 2018) - 18 percent nickel maraging steel bars and rods — specification	- Requirements of maraging steels with 18 percent nickel in the form of bars and rods of size less than 62.5 mm nominal diameter - Manufacturing process & chemical composition - Referenced documents and testing requirements

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Materials	ASTM A276 – 13a - Standard specification for stainless steel bars and shapes	- Scope and materials covered - Exclusions
	ASTM A314 – 19 - Standard specification for stainless steel billets and bars for forging	Specification stainless steel billets and bars for forging.
	ASTM A555/A555M – 20 - Standard specification for general Requirements for stainless steel Wire and Wire rods	- General requirements that shall apply to stainless wire and wire rods - Furnishing units
	GSO ASTM A276: 2009 - Standard specification For stainless steel bars and shapes	- Scope and materials covered - Exclusions
	GSO ISO 16143-2: 2013 - Stainless steels For general Purposes - Part 2: Semi-Finished products, bars, rods and sections	- Technical delivery details - Additional applicability of ISO 404 - Exclusions

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	IS 1570: Part 1: 1978 (Reaffirmed Year: 2019) (Part 5) - Schedules for wrought steels	- Applicable to steels with a high alloy content primarily interred for corrosion and heat-resisting Purposes - Chemical composition - Referenced documents
	IS 13177: 1991 (Reaffirmed Year: 2021) - Passivation of stainless steel Parts	- Specification for chemical dip passivation coating on wrought or stainless steels - References - Sampling and testing
	IS 13147: 1991 (Reaffirmed Year: 2021) - Black oxide coating on stainless steel parts	- General specifications and tests - Referenced documents

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	A276 / A276M – 17 - Standard specification for stainless steel bars and shapes	- Specification for hot-finished or cold-finished bars - Inclusions and Exclusions - Referenced documents
	ASTM A493 – 16 - Standard specification for austenitic ductile iron castings	- Austenitic ductile iron castings used for their resistance to heat, corrosion and wear - Exclusions - Referenced documents
	ASTM A582 - Standard specification for free-machining stainless steel bars	- Specifications for hot-finished or cold-finished bars - Exclusions - Referenced documents
	ASTM - A484 - Standard specification for general requirements for stainless steel bars, billets, and forgings	- General requirements applicable to wrought stainless steel bars, shapes, forgings, and billets or other semi-finished materials - Exclusions - Referenced documents
	AS 2837—1986 - Wrought alloy steels — Stainless steel bars and semi-finished products.	- Specifications for stainless steel bars and other semi-finished products - Referenced documents
	GSO ASTM - A484/A484M – 2016 - Standard specification for general requirements for stainless steel bars, billets, and forgings	- General requirements applicable to wrought stainless steel bars, shapes, forgings, and billets or other semi-finished materials - Exclusions - Referenced documents
	GSO ISO 1684 – 2013 - Wire, bar and tube drawing dies - specifications	- Main dimensions for tool types - Codified designation of dies - Referenced documents

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 3711: 2020 - Steel and steel products — location and preparation of samples and test pieces for mechanical testing	- Identification, location and preparation of samples for mechanical testing - General testing requirements - Documents referenced
	IS 13015: 1991 (Reaffirmed Year: 2018) - Steel products macro etch testing, inspection and ratings, specifications	- Significance and applications - Sample Selections - Test methods and referenced documents
	IS/ISO 14284: 1996 (Reaffirmed Year: 2021) - Steel and iron - Sampling and preparation of samples for the determination of chemical composition	- Methods of analysis - Requirements of sampling and preparation - Safety hazards - Documents referenced
	IS 15839: 2021 - Determination of ferrite content in austenitic stainless steel castings	- Significance of ferrite contents - Determination of chemical composition - Determination of ferrite by metallographic means
	ISO 15510: 2014 - Chemical composition of stainless steels	Chemical composition of stainless steels

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	EN ISO 3651: 2018 - Determination of resistance to intergranular corrosion of stainless steels Ferritic, austenitic and ferritic-austenitic (duplex) stainless steels. Corrosion test in media containing sulfuric acid	Determination of resistance to intergranular corrosion of stainless steels
	ASTM A380 - Standard practice for cleaning, descaling, and passivation of stainless steel parts, equipment, and systems	- Standard recommendations and precautions for cleaning, descaling, and passivating of new stainless steel parts, assemblies, equipment, and installed system - Referenced documents
	A800 / A800M – 20 -Standard practice for estimating ferrite content of stainless steel castings containing both ferrite and austenite	- Determination of chemical compositions - Determinations of magnetic responses - Referenced documents
	ASTM A751 – 20 - Standard test methods and practices for chemical analysis of steel products	- Test methodologies wet and instrumental - Referenced documents
	ISO 15510: 2014 - Chemical composition of stainless steels	Chemical composition of stainless steels

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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<https://www.legislation.gov.uk/ukxi/2012/2619/contents/made>
 - b. USA: <https://www.compliancegate.com/heavy-metal-regulations-united-states/>
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11. BIS Certified Labs for Testing products as per IS Standards: http://164.100.105.198:8096/bis_access/iswise_v2.html
12. Mandatory BIS standards: <https://www.bsbedge.com/mandatory-indian-standards>
13. QCO: <https://steel.gov.in/sites/default/files/145%20QCO%20Order.pdf>
14. ISO Standards: <https://www.iso.org/obp/ui#search>
15. International Standards:
 - a. EU: <https://www.en-standard.eu/>
 - b. USA: <https://www.en-standard.eu/astm-standards/>, <https://www.astm.org/>
 - c. AUS: <https://www.standards.org.au/>
 - d. GULF: <https://www.gso.org.sa/store/>
16. NABL accredited laboratories: <https://nabl-india.org/>, <https://parakh.ncog.gov.in>

STEEL WIRE, GHAZIABAD

1. INTRODUCTION

A wire is a usually a cylindrical, flexible strand or rod of metal. Wires are used to bear mechanical loads or conduct electricity and telecommunication signals. A Wire is commonly formed by drawing the metal through a hole in a die or draw plate. Wires come in various standard sizes, as expressed in terms of a gauge number. The term 'wire' is also used more loosely to refer to a bundle of such strands, as in "multi-stranded wire", which is more correctly termed a wire rope in mechanics, or a cable in electricity.



Wire comes in solid core, stranded, or braided forms. Although usually circular in cross-section, wires can be made in square, hexagonal, flattened rectangular or other cross-sections, either for decorative purposes or for technical purposes such as high-efficiency voice coils in loudspeakers. Edge-wound coil springs, such as the Slinky toy are made of special flattened wire.

Steel wire is an extremely useful and versatile material which finds uses in various

everyday objects across commercial, industrial and military applications for cleaning, spring-loaded devices, reinforcing, structural integrity and even for making music, etc. Steel wire is manufactured to different levels of strength, elasticity and thickness for different purposes. Steel wire can be broadly classified into eight different categories depending on their intended purposes.

Ghaziabad is well known for its industrial prowess and is home to a large number of units producing a wide range of products made from specialty long steel, tool steel and various alloy steels for the industry such as cold drawn steel wires, spring steel, carbon wire etc. The units cater to not only domestic but the international markets across industry segments such as automotive, textile, construction amongst a few.

2. RAW MATERIALS

The raw material/ feed wire for drawing steel wire are:

- Steel “wire rod” produced on modern hot rolling mills from billets issued of the “Electric furnace + AOD converter + continuous casting” process.

The wire rod comes in sizes 5.5 to 30 mm with 5.5 mm (0.218”) being the prevalent size; the wire rod is annealed and pickled or simply pickled if the rod is controlled-cooled.

3. MANUFACTURING PROCESS

Steel wires are manufactured in various methods depending on their size. The various methods are described as below:

3.1. Preparing the rod for drawing

Various methods are utilized for cleaning the rod of scales or the hard oxide layer. This is usually done in the following ways:

- a. Chemical reaction with dilute sulphuric or hydrochloric acid. This process is also called pickling
- b. Mechanical means such as shot blasting or by brushing

The pickling or chemical method is preferred by manufactures but the disadvantage associated with this method is that along with scale, the metal is also attacked by acid. The reaction releases nascent hydrogen some of which gets absorbed by the material of wire resulting in making the wire material brittle. Secondly, the wire surface may get pitted by reaction of acid which spoils the surface quality of drawn wire.

To counter this, now-a-days, inhibitors are added to the acid solution which cover up the newly formed metal surface after the scale gets peeled off and act as a barrier between the acid and the metal. The temperature of acid solution is raised up to 95°C in order to increase the rate of reaction. For mild steel wires,

dilute HCl or H₂SO₄ may be used. Generally, the waste acid of other industries is employed.

3.2. Dry drawing

For wire sizes > 0.8 mm, the drawing process is known as “dry drawing”. In this case, the wire rod is subjected to a phosphate pre-coat which provides a suitable substrate for wire drawing lubricants, typically Sodium (Na) or Calcium (Ca)-containing soaps.

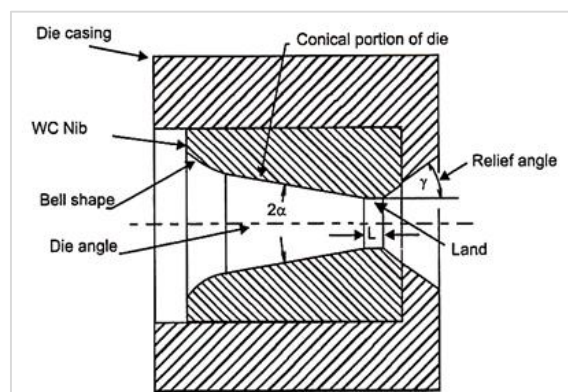


Figure 1: Cross Section of dies used in dry drawing

The stainless-steel wire is then drawn through a series of die-boxes containing the lubricant in powder form and the tungsten carbide die, often on so-called multi-holes or multiple-die machines (up to 12 in some of the most modern machines). Depending on the finish size and the grade, it may be necessary to do intermediate operations of in-line cleaning followed by in-line annealing in N₂/He reducing atmosphere and eventual re-coating for finish drawing. Depending on the range of

tensile strength and the surface condition specified, a final anneal and a “skin-pass” light draw might be performed. Also, single block drawing is sometimes used with grease as lubricant, to obtain bright surface finish.



Figure 2: Dry drawing

3.3. Wet drawing

The Dry method is not used for fine drawing. For fine wire production, typically < 0.80mm, the process is quite different and known as “wet drawing”. The feed wire is a so-called “redraw wire” in sizes smaller than 1.2 mm and fully annealed. The drawing process is performed through a series of diamond dies (up to 30 in some machines) fully immersed in an oil-type lubricant. This delicate process is followed for soft wire by in-line cleaning, annealing in N_2/H_2 atmosphere and precision spooling on special take-ups adapted to the various types of spools. Ultra-fine steel wires down to

about 0.010 mm are thus produced by a few highly specialized stainless steel wire producers.



Figure 3: Wet drawing

3.4. Die rotation

Rotation of die during wire drawing process helps in making the wear of die more uniform and hence increases die life.

Application of ultrasonic vibrations perpendicular to the axis of wire helps in reducing friction. However, the effect is rather small and the cost is high. Therefore, at present it is not economical due to high initial cost of the equipment and its maintenance.

3.5. Machinery

Large diameter wires and straight bars are generally drawn on draw benches. Small diameter wires which can be wound are drawn on block or drum type machinery. The rotation of circular drum or block on which the wire is wound pulls the wire through the die.

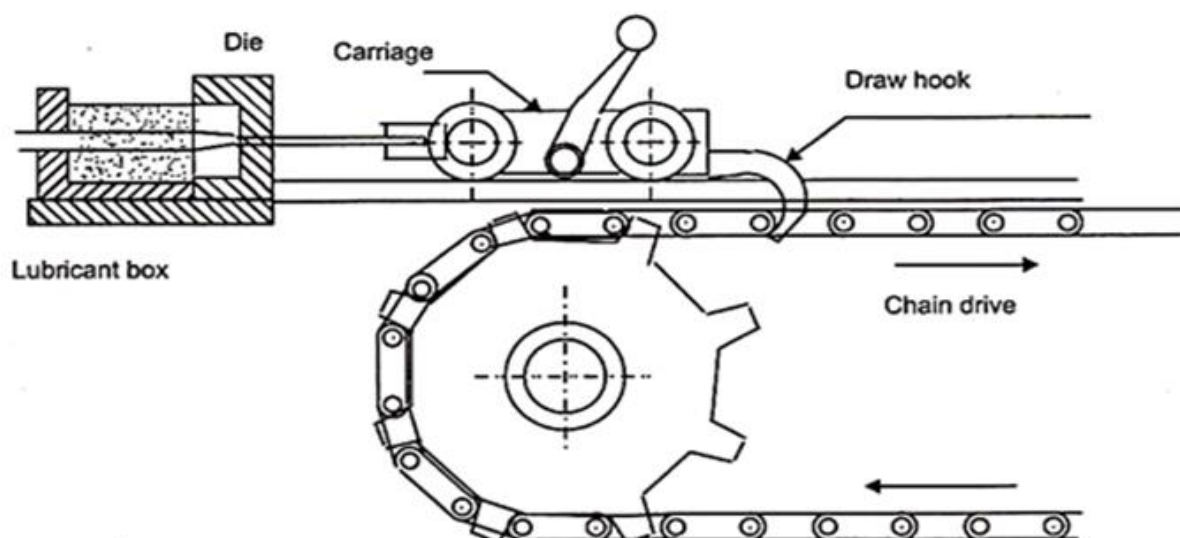


Figure 4 (a): Large Diameter Steel Wire Pulling

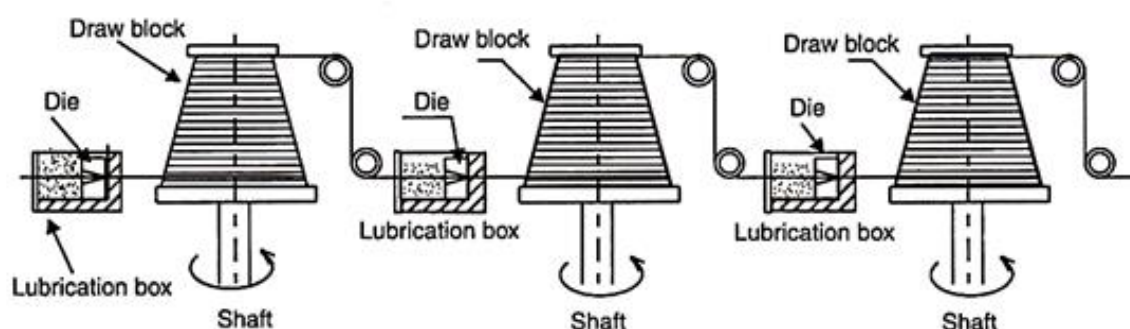


Figure 4 (b): Large Diameter Steel Wire Pulling

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Random checking of diameters
- Check for center - burst or cracking
- Check for abrasions and cuts

(some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international

- Federation of Indian Export Organizations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs

may be obtained from Department of Commerce

(<https://commerce.gov.in/international-trade/trade-agreements/>).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	a. Steel and steel products-Quality Control order SO 4637 (E)	a. BIS	a. Steel wires for mechanical springs

5.2. International

S. No.	Region/Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Restriction of Hazardous Substances (RoHS) Directive (EC)2015/863 b. Regulation (EU) 2021/854 of 27 May 2021	a. European Commission	a. Compliance on not having more than the prescribed limit for lead, chromium and hexavalent chromium b. Anti-dumping legislations on imports of stainless steel cold rolled flats originating in India and Indonesia
2	United States of America	a. California proposition 65, Act of 1986 b. Restriction of Hazardous Substances (RoHS) Regulations	a. California State b. Respective State Governments having RoHS regulations in place	a. Any manufacturing unit having more than ten employees are required to label products if they contain carcinogenic products b. Compliance on not having more than the prescribed limit for Lead, Chromium and Hexavalent chromium.

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
General (Refer Table 1)	IS 1956: Part 3: 2019	Glossary of Terms Relating to Iron and Steel Part 3 Long Products (Including Bars, Rods, Sections and Wires) (Second Revision)	European Union	BS EN 10218-2: 2012	Steel wire and wire products. general Wire dimensions and tolerances
	IS 8910: 2022	General technical delivery requirements for steel and steel products	USA	ASTM A227 / A227M - 17	Standard specification for steel Wire, Cold-Drawn for Mechanical Springs

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
	SP 17: 1979 (Reaffirmed Year: 2018)	Index to IS 1956, Glossary of terms relating to iron and steel	USA	ASTM A510 / A510M - 11	Standard specification for general Requirements for Wire rods and Coarse Round Wire, Carbon steel, and Alloy steel
	IS 1956: Part 1: 1976	Glossary of terms relating to iron and steel Part I General metallurgy, heat treatment and testing	-	-	-
	IS 1956: Part 5: 1976 (Reaffirmed Year: 2018)	Glossary of terms relating to iron and steel; Bright steel bar and steel wire	-	-	-
	SP 14: 1976 (Reaffirmed Year: 2018)	Index to steel designations	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	IS 280: 2006 (Reaffirmed Year: 2021)	Mild steel wire for general engineering purposes, Fourth revision	European Union	BS EN 10323: 2004	Steel wire and products. Bead wire
	IS 1673: 1984 (Reaffirmed Year: 2021)	Specification for mild steel wire cold heading quality	European Union	BS EN 10264-4: 2012	Steel wire and wire products. steel wire for ropes cold drawn non-alloy steel wire for ropes for general applications
	IS 1835: 1976 (Reaffirmed Year: 2022)	Specification for round steel wire for ropes	European Union	BS EN ISO 16120-2: 2017	Non-alloy steel wire rod for conversion to wire specific requirements for general purpose wire rod
	IS 2062: 2011 (Reaffirmed Year: 2021)	Hot rolled medium and high tensile structural steel	USA	ASTM A370 - 2	Standard test methods and definitions for mechanical testing of steel products
	IS 3975: 1999 (Reaffirmed Year: 2021)	Low carbon galvanized steel wires formed wires and tapes for armouring of cables – Specification Third Revision	Australia	ISO 22034-1: 2007	Steel wire and wire products
	IS 4224: 1972 (Reaffirmed Year: 2022)	Specification for steel wire for staples pins and clips First Revision	Middle East	GSO ASTM A276: 2009	Standard specification for stainless steel bars and shapes
	IS 8510: Part 1 to 3 1977	Specification for tinned steel wire for banding of armatures and rotors	USA	ASTM A853 - 19	Standard specification for steel Wire, Carbon, for general Use

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
	IS 4454: Part 1: 2001 (Reaffirmed Year: 2021)	Steel wires for mechanical springs - Part I: Patented and cold drawn steel wires - Unalloyed	-	-	-
	IS 6527: 1995 (Reaffirmed Year: 2019)	Specifications-stainless steel wire rods	-	-	-
Manufacturing/ Processing (Refer Table 3.a & 3.b)	IS 14652: 1999 (Reaffirmed Year: 2018)	18 percent nickel maraging steel bars and rods — specification	European Union	BS EN 10323: 2004	Steel wire and products. Bead wire
	IS 1570: Part 1: 1978 (Reaffirmed Year: 2019) (Part 5)	Schedules for wrought steels-Stainless and heat resistant steels	USA	ASTM A227/A227M-17	Standard specification for steel Wire, Cold-Drawn for Mechanical Springs
	IS 279: 1981 (Reaffirmed Year: 2021)	Specification for galvanized steel wire for telegraph and telephone purposes (Third Revision)	USA	ASTM - A484	Austenitic bar and forgings
	-	-	USA	ASTM A679 / A679M - 17	Standard specification for steel Wire, high tensile strength, cold drawn
	-	-	Middle East	GSO ISO 1684: 2013	Wire, bar and tube drawing dies - Specifications
	-	-	European Union	BS EN 10244-5: 2001	Steel wire and wire products. Non-ferrous metallic coatings on steel wire nickel coatings
Finishing (Refer Table 4.a & 4.b)	IS 3711: 2020	Steel and steel products- Location and preparation of samples and test pieces For mechanical testing	European Union	BS EN 10218-2: 2012	Steel wire and wire products. General test methods
	IS 13015: 1991 (Reaffirmed Year: 2018)	Steel products- macro etch testing, specifications for inspection and rating	USA	ASTM A380	Substantial contamination, heat scale or heat tinting through passivation of pickling
	IS/ISO 14284: 1996 (Reaffirmed Year: 2021)	Steel and iron - Sampling and preparation of samples for the determination of Chemical composition [MTD 4: wrought steel products]	USA	ASTM A751 - 20	Standard test methods and practices for chemical analysis of steel products

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
	IS 15839: 2021	Determination of ferrite content in austenitic stainless steel castings [MTD 22: Metallography and Heat treatment]	Australia	ISO 15510: 2014	Stainless steels - Chemical composition
	IS 228: Part 1: 1987 (Reaffirmed Year: 2018)	Methods of chemical analysis of steels: Part 1 Determination of carbon by volumetric method	-	-	-
	IS 10206: 1982 (Reaffirmed Year: 2018)	Methods of sampling steel wires	-	-	-

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	IS 1956: Part 3: 2019 - Glossary of Terms Relating to Iron and Steel Part 3 Long Products (Including Bars, Rods, Sections and Wires) (Second Revision)	Glossary of Terms Relating to iron and steel, Long products (Including bars, rods, sections, wires) Illustrations of typical sheet pilings
	IS 8910: 2022 - General technical delivery requirements for steel and steel products	General technical delivery requirements for all steel products covered by ISO 6929 Exclusions
	SP 17: 1979 (Reaffirmed Year: 2018) - Index to IS 1956, Glossary of terms relating to iron and steel	Glossary of terms, giving clause and part numbers of IS 1956
	IS 1956: Part 1: 1976 - Glossary of terms relating to iron and steel Part I General metallurgy, heat treatment and testing	Glossary of terms used in metallurgy, heat treatment and testing
	IS 1956: Part 5: 1976 (Reaffirmed Year: 2018) - Glossary of terms relating to iron and steel, bright steel bar and steel wire	Glossary of terms
	SP 14: 1976 (Reaffirmed Year: 2018) - Index to IS 1570: Part 1: 1978 (Reaffirmed Year: 2019) steel Designations	Glossary of terms
	BS EN 10218-2 – 2012 - Steel wire and wire products. general wire dimensions and tolerances	Steel wire and wire products. general wire dimensions and tolerances
	ASTM A510 / A510M – 11 - Standard specification for general requirements for wire rods and coarse round wire, carbon steel, and alloy steel	Specifications for general requirements for wire rods and coarse round wire, carbon steel, and alloy steel
	ASTM A227 / A227M – 17 - Standard specification for steel Wire, Cold-Drawn for Mechanical Springs	Classes of round cold-drawn steel spring wire having properties and quality for the manufacture of mechanical springs that are not subject to high stress

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	IS 280: 2006 (Reaffirmed Year: 2021) - Mild steel wire for general engineering purposes	- Mild steel wire for sizes 0.125 to 12.5 mm for general engineering purposes - Chemical and mechanical properties - Manufacturing and testing requirements - Tolerances
	IS 1673: 1984 (Reaffirmed Year: 2021) - Specification for mild steel wire, cold heading quality	- Requirements for two grades of cold drawn mild steel wires ranging from 1.60 to 12.50 mm - Terminology - Chemical and mechanical properties - Testing requirements
	IS 1835: 1976 (Reaffirmed Year: 2022) - Specification for round steel wire for ropes	- Grades of wires requirements, for use in wire ropes - Terminology and types - Chemical and mechanical properties - Testing requirements
	IS 2062: 2011 (Reaffirmed Year: 2021) - Hot rolled medium and high tensile structural steel — Specification	- Requirements of steel including micro-alloyed steel plates, strips, shapes and sections (angles, tees, beams, channels, etc.), flats, bars, etc., for use in structural work - Grades - Referenced documents
	IS 3975: 1999 (Reaffirmed Year: 2021) - Low carbon galvanized steel wires, formed wires and tapes for armouring of cables - Specification	- specifications - Chemical and Mechanical properties - Tolerances - Sampling criteria
	IS 4224: 1972 (Reaffirmed Year: 2022) - Steel wire for staple pins and clips	- Chemical composition - Sizes, Tolerances and finishes
	IS 4454: Part 1: 2001 (Reaffirmed Year: 2021) - Steel wires for mechanical springs - Part I: Patented and cold drawn steel wires - Unalloyed	- Steel specification for manufacture of springs for static and dynamic duty - Referenced documents - Test Methodology
	IS 6527: 1995 (Reaffirmed Year: 2019) - Stainless steel wire rods, specifications	- Requirements for stainless steel wire rods - Exclusions & referenced documents - Chemical compositions and freedom from defects
	IS 8510 - Part 1 to 3 (1977): Specification for tinned steel wire for banding of armatures and rotors	- Grades - Sizes and Tolerances - Sampling and tests

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	BS EN 10323 – 2004 - Steel wire and products. Bead wire	Steel wire and products. Bead wire
	BS EN 10264-4 – 2012 - Steel wire and wire products. Steel wire for ropes stainless steel wire	Steel wire and wire products. steel wire for ropes stainless steel wire
	BS EN 10264-2 – 2012 - Steel wire and wire products. Steel wire for ropes cold drawn non-alloy steel wire for ropes for general applications	Steel wire for ropes Cold drawn non-alloy steel wire for ropes for general applications

Stage	Code	Scope
	BS EN ISO 16120-2 – 2017 - Non-alloy steel wire rod for conversion to wire Specific requirements for general purpose wire rod	- Non-alloy steel wire rod for conversion to wire - Specific requirements for general purpose wire rod
	ASTM A370 – 2 - Standard Test Methods and Definitions for Mechanical Testing of steel products	Determination of mechanical properties of steel
	ASTM A853-19 - <i>Standard specification for steel Wire, Carbon, for general Use</i>	Carbon steel wire, supplied in coils, for general use. Produced hard drawn, annealed in process, or annealed at finish size.
	ISO 22034-1 – 2007 - Steel wire and wire products	Steel wire and products
	GSO ASTM A276: 2009 - Standard specification For stainless steel bars and shapes	Specification for stainless steel bars and shapes

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	IS 14652: 1999 (Reaffirmed Year: 2018) - 18 percent nickel maraging steel bars and rods — specification	- Product Analysis - Mechanical requirements and analysis
	IS 1570: Part 1: 1978 (Reaffirmed Year: 2019) (Part 5) - Schedules for wrought steels	- Designation of steel on mechanical properties - Documents referenced
	IS 279: 1981 (Reaffirmed Year: 2021) - Galvanized steel wire for telegraph and telephone purposes	- Terminology - Chemical and mechanical properties

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	BS EN 10323 – 2004 - Steel wire and products. Bead wire	Steel wire, wire ropes and link chains
	BS EN 10244-5 – 2001 - Steel wire and wire products. Non-ferrous metallic coatings on steel wire nickel coatings	Steel wire and wire products. Non-ferrous metallic coatings on steel wire nickel coatings
	ASTM A493 – 16 - Standard specification for Austenitic Ductile iron Castings	- Austenitic ductile iron castings used for their resistance to heat, corrosion and wear - Exclusions - Referenced documents
	ASTM A582 - Standard specification for Free-Machining stainless steel bars	- Specifications for hot-finished or cold-finished bars - Exclusions - Referenced documents
	ASTM A679 / A679M – 17 - Standard specification for steel wire, high tensile strength, cold drawn	Uncoated, high tensile strength, cold drawn steel spring wire
	GSO ISO 1684: 2013 - Wire, bar and tube drawing dies - specifications	- Main dimensions for tool types - Codified designation of dies - Referenced documents

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 3711: 2020 -Steel and steel products — location and preparation of samples and test pieces for mechanical testing	- Identification, location and preparation of samples for mechanical testing - general testing requirements - Documents referenced
	IS 13015: 1991 (Reaffirmed Year: 2018) - Steel products macro etch testing, inspection and ratings, specifications	- Significance and applications - Sample Selections - Test methods and referenced documents
	IS/ISO 14284: 1996 (Reaffirmed Year: 2021) - Steel and iron - Sampling and Preparation of Samples for the Determination of Chemical Composition	- Methods of analysis - Requirements of sampling and preparation - Safety Hazards - Documents referenced
	IS 15839: 2021 - Determination of Ferrite Content in Austenitic stainless steel Castings	- Significance of Ferrite contents - Determination of chemical composition - Determination of ferrite by metallographic means
	ISO 15510: 2014 - Chemical composition of stainless steels	Chemical composition of stainless steels
	IS 10206: 1982 (Reaffirmed Year: 2018) - Chemical and mechanical characteristics	- Terminology - Chemical and mechanical characteristics - Inspections

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	BS EN 10218-2 – 2012 - Steel wire and wire products.	Steel wire and wire products. general wire dimensions and Tolerances
	ASTM A380 - Standard Practice for cleaning, descaling, and passivation of stainless steel parts, equipment, and systems	- Standard recommendations and precautions for cleaning, descaling, and passivating of new stainless steel parts, assemblies, equipment, and installed system - Referenced documents
	ASTM A751 – 20 - Standard test methods and practices for chemical analysis of steel products	- Test methodologies wet and instrumental - Referenced documents
	ISO 15510: 2014 - Stainless steels - Chemical composition	- Stainless steels - Chemical composition - Applicable to all wrought product forms including ingots and semi-finished material.

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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 - b. <https://www.legislation.gov.uk/ukxi/2012/2619/contents/made>
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 - d. <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/CFRSearch.cfm?CFRPart=111&showFR=1&subpartNode=21:2.0.1.1.11.4>
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 - a. EU: <https://www.en-standard.eu/>
 - b. USA: <https://www.en-standard.eu/astm-standards/>, <https://www.astm.org/>
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OIL FILTER, GHAZIABAD

1. INTRODUCTION

An Oil filter is usually a cylindrical tube designed to remove particulate contaminants detrimental to an engine from the many oils (e.g., such as engine oil, transmission oil, lubricating oil, or hydraulic oil) used to ensure smooth operation. Their chief use is in the internal-combustion engines for motor vehicles, powered aircraft, railway locomotives, ships and boats and static engines such as generators and pumps. Other vehicle hydraulic systems such as those in automatic transmissions and power steering are often equipped with an oil filter. Gas turbine engines such as those on jet aircraft also require the use of oil filters.



The oil filter helps remove contaminants from the engine's oil that can accumulate over time as the oil keeps the engine clean. If left unfiltered for a period of time, it could become saturated with tiny, hard particles that can wear surfaces in the engine.

Ghaziabad district is well known for its industrial prowess and is home to a large number of units producing a wide range of

automobile components and other associated engineering goods. The units cater to not only domestic but the international markets across industry segments such as motor vehicles, railway locomotives and aircrafts. Modern day oil filters are broadly of two main categories, namely, the Full Flow System (In-line oil filter) and the Bypass systems

1.1. Full Flow System

A full-flow system comprises of a pump which sends pressurized oil through a filter to the engine bearings after which the oil returns by gravity to the sump which is also called as the oil pan. The function of the full-flow filter is to protect the engine from wear through abrasion.

1.2. By-pass System

A By-pass system is slightly more elaborate and complicated system than that of the full flow system. Modern by-pass oil filter systems are secondary systems whereby the main purpose of the bypass is to have a secondary filtration system to keep the oil in good condition, keep it free of dirt, soot and water as well as prevent oil starvation. They are often used in conjunction with a full flow filter. Bypass systems were originally used on commercial and industrial diesel engines with large oil capacities where the cost of oil analysis testing and extra filtration to

extend oil change intervals made economic sense but they are becoming more common in private consumer applications such as automobiles.

2. RAW MATERIALS

The raw material for manufacturing an Oil filter are:

- Steel Sheet for stamped products
- Springs
- Rubber Sealing Rings
- Anti-Drain Valves
- Support tube fabricated from meshed sheet of Steel
- Filter Media

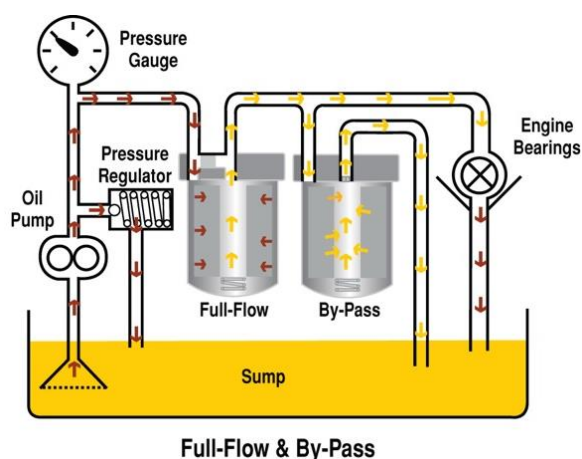


Figure 1: Working of a Filter

3. MANUFACTURING PROCESS

Following are the key steps in the manufacture of an oil filter:

3.1. Sheet Steel Cutting

The first step in the production of an oil filter is cutting of the steel sheet. As the steel sheet passes through the cutting machine, specific sizes that are required for the filter-making

process are cut. The threaded plates that form the thickest parts are cut from a 2mm thick sheet while the tapping plate, outer casing and the upper-end cover are cut from a sheet of thickness between 0.25 mm-0.5 mm.

3.2. Metal Stamping

The stamping process involves steel sheet blanking, extraction of the outer casing, stamping of the base plate and the end cap of the filter element. Most of the metallic parts of the filter are made at this stage of the filter making process. This process is also used to shape the outer shell canister and the cover plate. The cover plate comprises of a tapping plate and the cover plate. The tapping plate is first threaded and then the cover plate and tapping plate are spot welded together.



Figure 2: Exploded view of Oil Filter

3.3. Installation of the Bypass Valve

The bypass valve is riveted to the bottom end caps of the filter element. For this, the die, spring and the bottom end caps are placed in

the automatic riveting machine. The machine rivets the lower end cap assembly.

3.4. Fabricating the Support Tube

Making the support tube starts with cutting the sheet metal into the specified size after which another machine perforates the cut tin-plated steel sheets and rolls them into tubes. These are the finished center tubes and are ready for the next stage of the filter making process where it will be inserted in the core.

3.5. Oil Filter element

The filter paper is cut to a size that can fit inside the canister. Shaping the width, height and number of folds are done using the folding machine and then the pleating machine automatically pleats the prepared paper. After pleating, the paper ribbons are cut into pre-defined lengths and folded into a cylindrical shape. The center tube/support tube is placed within the paper core and fastened to the filter element.

3.6. Gluing the End Caps

An automated glue gun applies adhesive to the inner sides of the end caps and they are stuck manually to the filter element.

3.7. Assembly

First, the base plate is loaded into the assembly line and the anti-drain back valve is installed after which the filter element is placed on the base plate. The support spring is put

on the filter element core, followed by the filter canister on the filter assembly. Finally, using a seamer machine, the filter edges are folded down to form a rim.

3.8. Testing

The test phase begins when the filter is assembled to ensure the quality and conformance to the specifications and requirements before packaging and dispatch.

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- The finished outer casings should not have any dents or paint peeling off
- The metal components should be free from rust
- The Filters after testing and drying should be carefully inspected for any signs of moisture

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organizations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade

Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (<https://commerce.gov.in/international-trade/trade-agreements/>).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	a. Steel and steel products-Quality Control order SO 4637 (E)	a. BIS	a. Steel wires for mechanical springs

5.2. International

S. No.	Region/Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Regulation (EU) 2021/646	a. European Commission	a. Directives and regulations on motor vehicles, their trailers, systems and components
2	United States of America	a. Federal Register, Vol. 64, No. 234, pp. 68546-68573 on December 7, 1999. b. NAICS 336, 4231, 8111	a. United States Environment Protection Agency	a. Motor Vehicles waste disposal wells

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
General (Refer Table 1)	IS 3233: 1965 (Reaffirmed Year: 2018)	Glossary of terms for safety and relief valves and their parts		ISO 11841 – 1: 2000	Road Vehicles and Internal combustion Engines, Filter vocabulary-1 Definition of Filter and Filter components
	IS 1956: Part 5: 1976 (Reaffirmed Year: 2018)	Glossary of Terms Relating to Iron and Steel:		ISO 11841 – Part 2: 2000	Road Vehicles and Internal combustion Engines, Filter vocabulary-2 Definition of

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
		Part 5 Bright steel bar and steel wire			Filter and Filter components
	-	-		ISO 1219-1: 2012	Fluid power systems and components-Graphic symbols and Circuit Diagrams- Part 1,Graphic Symbols
	-	-	USA	ASTM D7720 -11 (2017)	Standard Guide for Statistically Evaluating Measured Alarm Limits when Using Oil Analysis to Monitor Equipment and Oil for Fitness and Contamination
Raw Materials (Refer Table 2.a & 2.b)	IS 2507: 1975 (Reaffirmed Year: 2021)	Specification for cold-rolled steel strips for springs	USA	ASTM A1000/A1000M -17	Standard Specification for Steel Wire, Carbon and Alloy Specialty Spring Quality
	IS 11149: 1984 (Reaffirmed Year: 2019)	Specifications for rubber gaskets	-	-	-
Manufacturing/ Processing (Refer Table 3)	-	-	Australia	ISO 7747: 1983	Road vehicles -- Filter elements for full flow oil filters -- Dimensions
	-	-	Australia	ISO 4548-6: 2021	Methods of test for full-flow lubricating oil filters for internal combustion engines — Part 6: Static burst pressure test
	-	-	USA	ASTM B79 3-16(2021)	Standard specification for zinc casting alloy ingot for sheet metal forming dies and plastic injection moulds
Finishing (Refer Table 4.a & 4.b)	IS 11509: Part 1: 2012 (Reaffirmed Year: 2017)	Methods of tests for full flow lubricating oil filters for Internal combustion engines Part 1: Differential pressure/Flow characteristics	USA	ASTM D2068 - 20	Standard Test Method for Determining Filter Blocking Tendency
	IS 11509: Part 2: 2012 (Reaffirmed Year: 2017)	Methods of tests for full flow lubricating oil filters for Internal combustion engines Part 1: Differential	USA	ASTM D7898 - 14(2020)	Standard Practice for Lubrication and Hydraulic Filter Debris Analysis (FDA) for Condition Monitoring of Machinery

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
		pressure/Flow characteristics			
	-	-	USA	ASTM F316 - 03(2019)	Standard Test Methods for Pore Size Characteristics of Membrane Filters by Bubble Point and Mean Flow Pore Test

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	ISO 11841 – 1: 2000 - Road Vehicles and Internal combustion Engines, Filter vocabulary-1, Definition of Filter and Filter components	- Definition of filters and filter components - Defines terms for filters and components used in road vehicles and internal combustion engines for general use (e.g. marine engines, stationary engines)
	ISO 11841 – Part 2: 2000 - Road Vehicles and Internal combustion Engines, Filter vocabulary-2 Definition of Filter and Filter components	- Definitions of characteristics of filters and their components - Defines terms for the characteristics of filters and filter components used in road vehicles and internal combustion engines for general use (e.g. marine engines, stationary engines)
	ASTM D7720 -11 (2017): Standard Guide for Statistically Evaluating Measured Alarm Limits when Using Oil Analysis to Monitor Equipment and Oil for Fitness and Contamination	- Requirements to statistically evaluate measure and alarm thresholds, which are called alarm limits - A basic statistical process control technique to evaluate alarm limits when measured - Exclusions
	IS 1956: Part 5: 1976 (Reaffirmed Year: 2018) - Glossary of terms relating to iron and steel; Bright steel bar and steel wire	Glossary of terms

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	IS 11149: 1984 (Reaffirmed Year: 2019) - Specifications for rubber gaskets	- Requirements for vulcanized rubber gaskets for various applications - Test methods
	IS 2507: 1975 (Reaffirmed Year: 2021) - Specification for cold-rolled steel strips for springs	- Supply of Materials, conditions - Grades and chemical composition - Thickness and tolerances

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	ASTM A1000/A1000M-17 - Standard Specification for Steel Wire, Carbon and Alloy Specialty Spring Quality	- This specification covers round, shaped and flat rolled plain carbon and alloy steel spring wires, uniform in quality and temper, intended for the manufacture of mechanical springs that can withstand moderate fatigue stresses over some relatively low number of cycles. - Normative references

6.4. Table 3: (International Standards for Processing)

Stage	Code	Scope
Processing	ISO 7747: 1983 - Road vehicles -- Filter elements for full flow oil filters -- Dimensions	- Specifies the essential dimensional characteristics necessary for the interchangeability of filter elements - Dimensions recommendations for future manufacturing
	ISO 4548-6: 2021 - Methods of test for full-flow lubricating oil filters for internal combustion engines — Part 6: Static burst pressure test	- Method of testing full-flow lubricating oil filters for internal combustion engines to determine their ability to withstand: - Static pressure objective - Determination of bursting pressure and failure mode
	ASTM B793-16(2021) - Standard specification for zinc casting alloy ingot for sheet metal forming dies and plastic injection moulds	Specification of Zinc composites intended for use in the fabrication of dies for sheet metal stamping under drop hammer and hydraulic pressure

6.5. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 11509: Part 1: 2012 (Reaffirmed Year: 2017) - Methods of tests for full flow lubricating oil filters for Internal combustion engines Part 1, Differential pressure/Flow characteristics	- Diagrammatic representation of test rig - Types of Filters - Filter mounting and specifications for test liquids - Test methodology and sample test report format
	IS 11509: Part 2: 2012 (Reaffirmed Year: 2017) - Methods of tests for full flow lubricating oil filters for Internal combustion engines Part 1, Differential pressure/Flow characteristics	- Tests for determining the element by pass valve characteristics of full flow lubricating oil filters - Operational characteristics to be tested - Filter to be tested and diagrammatic representation of test rig

6.6. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	ASTM D2068 – 20 - Standard Test Method for Determining Filter Blocking Tendency	- Evaluation of the cleanliness of middle distillate fuels, and biodiesel and biodiesel blends for specifications and quality control purposes - Evaluation of Filter Blocking tendency (FBT) of middle distillate fuels - Exclusions and System of Units for measurements
	ASTM D7898 - 14(2020) - Standard Practice for Lubrication and Hydraulic Filter Debris Analysis (FDA) for Condition Monitoring of Machinery	- Cover the extraction, analysis, and information management pertaining to visible wear debris collected from oil system filters or debris retention screens - Exclusions

Stage	Code	Scope
	ASTM F316 - 03(2019) - Standard Test Methods for Pore Size Characteristics of Membrane Filters by Bubble Point and Mean Flow Pore Test	- Tests for determination of maximum pore size of a filter - Determination of the effect of various processes such as filtration, coating, or autoclaving on the maximum pore size of a membrane

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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 - b. USA: <https://www.en-standard.eu/astm-standards/>, <https://www.astm.org/>
 - c. AUS: <https://www.standards.org.au/>
 - d. GULF: <https://www.gso.org.sa/store/>
14. NABL accredited laboratories: <https://nabl-india.org/>, <https://parakh.ncog.gov.in>

PISTON, GHAZIABAD

1. INTRODUCTION

A piston is a component of a reciprocating engine, reciprocating pump, gas compressor, hydraulic cylinder and pneumatic cylinders, among various other similar mechanisms. Pistons are at the very heart of the reciprocating internal combustion engine, which is why they are often called "piston engines". At its most basic form, a piston is simply a solid cylinder of metal which moves up and down in the hollow cylinder of the engine block. The piston itself is slightly smaller than the cylinder cavity it moves in but it has piston rings under tension to achieve a nearly air-tight seal once it is installed within the engine cylinder. The piston is attached to a connecting rod which in turn is connected to the crankshaft and together they turn the reciprocating motion into rotational motion to drive the wheels.



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cater to not only domestic but the international markets across different industry segments.

In order to understand the working of a piston, it is imperative to understand the working of a common internal combustion engine. For this purpose, a simple four stroke internal combustion engine is being taken as a point of reference.

1.1. Four Stroke Internal Combustion Engine

- a. The piston is positioned inside a cylinder into which a mixture of air and fuel or air is introduced depending on the engine i.e. petrol or diesel.
- b. The fuel air mixture is compressed and ignited by a spark plug in case of a petrol engine or diesel being injected directly in case of a diesel engine.
- c. The hot gases expand, pushing the piston to the bottom of the cylinder. This position is also known as the Bottom Dead Centre (BDC) where the piston forms the largest volume in the cylinder. A connected flywheel returns the piston to the Top Dead center.
- d. The expanded or exhaust gases are expelled in the above-mentioned stroke

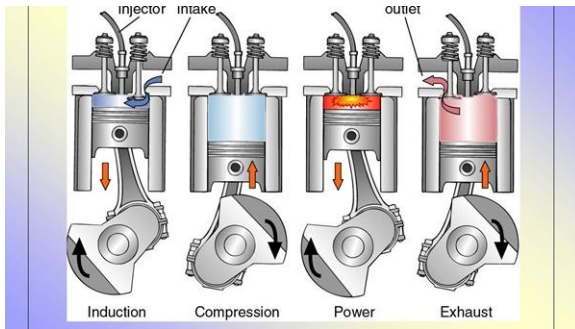


Figure 1: Working of an Internal Combustion Engine

The pressures and temperatures generated due to these actions can be up to 20 MPa (Mega Pascal Pressure Unit) and 450°C. In addition, a seal is made between the sliding piston and the walls of the cylinder so that the high-pressure gas above the piston does not leak past it and reduce the efficiency of the engine. This seal is usually provided by one or more piston rings. These are rings made of a hard metal and are sprung into a circular groove in the piston head. The rings fit closely in the groove and press lightly against the cylinder wall to form a seal and more heavily when higher combustion pressure moves around to their inner surfaces.

1.2. Components of a Piston

a. Piston Compression Rings: In modern engines, there are two or three compression rings fitted into the top grooves. The number of compression rings tends to improve the compression ratio. An oil control ring is fitted into the lower groove of the piston. Generally, the second and third

compression rings are taper-faced and supplied to improve oil sealing. To prevent rapid wear of these rings, they are usually coated with graphite, iron oxide and phosphate.

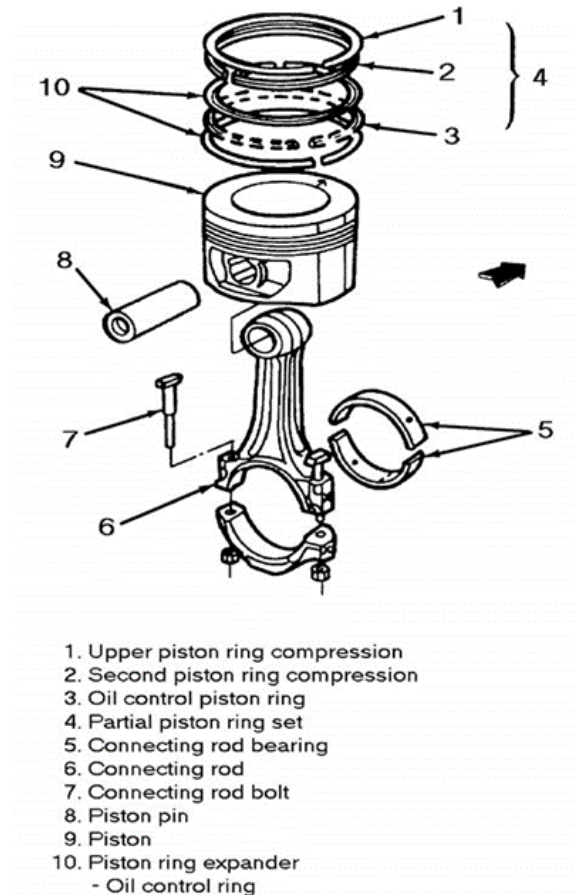


Figure 2: Exploded view of a Piston

b. Oil Control Piston Rings: Oil control rings have a series of slots which transfer excess oil through holes in the piston groove to the inside of the piston which is then returned to the sumps but leave sufficient oil to lubricate the cylinder walls. Oil control rings are subjected to the highest inherent pressure amongst the three rings installed in a piston. This is a critical

component as without it, oil would enter the cylinder and get burnt along with the fuel air mixture leading to increased oil consumption and lead to fouling the spark plug.

- c. Piston Ring Expander:** Multi-piece steel oil control rings usually consist of two thin steel rings (called rails) with a spacer-expander spring between them to keep the two rails apart and provide radial pressure. The piston rings are usually made of fine-grained alloyed cast iron. The piston rings are split so that they can be expanded and slipped over the piston head and into the recessed grooves cut in the piston. The rings usually have butt joints, however, in some heavy-duty engines, the joint may be angled, lapped or of the sealed type.
- d. Connecting Rod Bearing:** The large-end bearing of the crosshead connecting rod is often made separately and then the connecting rod bolt is fixed to the shaft and the antifriction alloy is often directly cast on the inner surface. For large-end bearings without cross-head connecting rods, the upper half of the connecting rod bearing is often made integral with the shaft while the lower half of the

connecting rod bearing cover is made separately.

- e. Connecting Rod and Bolt:** The connecting rod is defined as it is a mechanical component that helps to connect the piston and crankshaft of the engine. The two bolts prevent the crankshaft from yanking the cap off the end of the connecting rod dozens of times a second.
- f. Piston:** At its most basic, the piston is simply a solid cylinder of metal which moves up and down in the hollow cylinder of the engine block. The piston itself is slightly smaller than the hole it moves in but it has piston rings under tension to achieve an (nearly) air-tight seal once it is installed within the engine cylinder. The piston is attached via a wrist pin to a connecting rod which in turn is connected to the crankshaft and together they turn the up and down (reciprocating) motion into rotational motion to drive the wheel.
- g. Piston Pin:** Also referred to as a gudgeon pin, it connects the piston to the connecting rod and provides a bearing for the connecting rod to pivot upon as the piston moves. The gudgeon pin is typically a forged short hollow rod made of a steel alloy

of high strength and hardness that may be physically separated from both the connecting rod and piston or crosshead. The gudgeon pin has to operate under some of the highest temperatures experienced in the engine with difficulties in lubrication due to its location, while remaining small and light so as to fit into the piston diameter and not unduly add to the reciprocating mass.

2. RAW MATERIALS

The raw materials for manufacturing a piston are:

- Aluminum alloys in required grades
- Piston Compression Rings made from Cast Iron
- Oil control Piston Rings
- Graphite
- Tin
- Connecting pins and bolts

The piston and its connecting components operate under extreme temperature and pressure conditions and thus are amongst the most carefully designed components of an automobile and are subjected to vigorous quality checks with negligible deviations.

3. MANUFACTURING PROCESS

The following are the key steps in the manufacture of a Piston:

3.1. Designing

A piston's performance depends on its design and shape. Modern Internal combustion engines rendered steel obsolete as a primary material. Aluminum was found out to be a better alternative but as it conducts much heat, pistons are designed with a taper. The tapered shape allows the piston to expand as heat is applied so the piston does not bind in the cylinder bore.

3.2. Casting

In this method, the metal is melted and poured into a cast. They enable bulk production and are produced in sizes close to factory specifications. The strength of the produced piston and its components are comparatively lower than the forged parts.

3.3. Forging

In this method, pistons are mechanically shaped into a piston shape. They are hammered into a mold, forming the piston and by utilizing state-of-the-art CNC machines, most manufacturers can maintain exact specifications and tolerances. This makes them structurally more durable. Forged pistons are well known in the racing and performance industry because of their ability to withstand more heat, higher rpm, higher boost and higher temperatures. Another advantage to a forged piston is that

they can be individually made to any custom size or specification without the high cost of casting.

3.4. Lathing and Machining

A lathe machine is used to cut metal from the basic form of the piston, taking it closer to its finished shape. Tiny holes are then drilled into the sides to create the oil passages for the piston. The same lathe then impresses three rings into the top of the piston. These rings or grooves help the piston glide and allow it to form an air-tight seal and a large hole is then drilled through both sides of the piston. This is where the wrist pin will go which is used to attach the piston to the connecting rod during engine assembly.

3.5. Milling and Finishing

Metal from each side of the piston is removed to reduce weight after which the piston is subjected to grinding to smoothen all rough edges. The manufactured pistons are sprayed with de-ionized water to remove any traces of lubricants or oil.

3.6. Assembly

All the components are assembled manually and inspected for defects and subjected to various quality checks prior to being packaged for transport

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- The piston should not have any residue adhering to it
- Check the piston diameter for dimensional accuracy
- Check for the finish

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organizations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce

(<https://commerce.gov.in/international-trade/trade-agreements/>).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	NA	NA	NA

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
General (Refer Table 1)	IS 5791: 2006 (Reaffirmed Year: 2020)	Internal Combustion Engines - Piston Rings - Material Specifications	Australia	ISO 18669 - 1: 2021	Internal Combustion engines-Piston Pins, General Specifications
	IS/ISO 6621: Part 1: 2018	Internal Combustion Engines Piston Rings Part 1 Vocabulary (First Revision)	Australia	ISO 6625: 1986	Internal combustion engines — Piston rings — Oil control rings
	IS 8422	Specifications for piston rings for Internal Combustion Engine	-	-	-
	IS 1387: 1993 (Reaffirmed Year: 2018)	General requirements for the supply of metallurgical materials	-	-	-
	IS 1956: Part 1: 1976 (Reaffirmed Year: 2018)	Glossary of terms relating to iron and steel Part I General metallurgy, heat treatment and testing	-	-	-
	IS/ISO 6621: Part 2: 2003	Internal Combustion Engines - Piston Rings Part 2 Inspection Measuring Principles	-	-	-
	IS 5052: 1993 (Reaffirmed Year: 2022)	Aluminum and its alloys - temper designations	-	-	-
	IS 5047: Part 1: 1986 (Reaffirmed Year: 2022)	Glossary of terms relating to aluminum and aluminum alloys: Part 1 Unwrought and wrought metals	-	-	-
	IS 5047: Part 2: 1979 (Reaffirmed Year: 2022)	Glossary of terms relating to aluminum and aluminum alloys: Part 2 Plant and operations, thermal	-	-	-

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
		treatment, control and testing, finishing			
	IS 10259: 1982 (Reaffirmed Year: 2020)	General conditions for delivery and inspection of aluminum and aluminum alloy products	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	IS 7793: 1975 (Reaffirmed Year: 2022)	Specification of Aluminum Alloys for I.C Engines	Middle East	GSO ISO 2107: 2008	Aluminum and Aluminum Alloys - Wrought Products - Temper Designations
	IS 11952: 1986 (Reaffirmed Year: 2019)	Specification for Steels for Piston Pins (Gudgeon Pins)		ISO 3547-4: 2017	Plain bearings — Wrapped bushes — Part 4: Materials
	IS 11998: 1987 (Reaffirmed Year: 2022)	Specification for Aluminum for the manufacture of alloys for I.C engine Pistons	-	-	-
	IS 1820: 1979 (Reaffirmed Year: 2022)	Recommended shapes, Sizes and mass of aluminum notched bars for remelting purposes	-	-	-
Processing (Refer Table 3.a & 3.b)	IS 734: 1975 (Reaffirmed Year: 2022)	Specification for wrought aluminum and aluminum alloy forging stock and forgings (For General Engineering purposes)		ISO 16232: 2018	Road vehicles — Cleanliness of components and systems
	IS 8422: Part 1: 1997 (Reaffirmed Year: 2019)	Specification for piston rings for IC engines part 1 plain compression rings from 30 up to 200 mm nominal diameter r - rings		ISO 3547-1: 2018	Plain bearings — Wrapped bushes — Part 1: Dimensions
	IS 8422: Part 2: 1997 (Reaffirmed Year: 2019)	Piston Rings for IC Engines - Part II: Taper Faced Compression Rings from 30 up to 200 mm Nominal Diameter M-rings		ISO 4384-1: 2019	Plain bearings — Hardness testing of bearing metals — Part 1: Multilayer bearings materials
	IS 8422: Part 3: 1997 (Reaffirmed Year: 2019)	Piston Rings for IC Engines - Part III: Keystone Rings from 82 up to 200 mm Nominal Diameter T-rings 15Å	Australia	ISO 18669 - 2: 2020	Internal combustion engines — Piston pins — Part 2: Inspection measuring principles

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
	IS 8422: Part 4: 1997 (Reaffirmed Year: 2019)	Piston rings for IC engines: Part 4 Napier oil scraper rings from 30 up to 200 mm nominal diameter N-Rings	-	-	-
	IS 8422: Part 5: 1997 (Reaffirmed Year: 2019)	Piston rings for IC engines: Part 5 Stepped oil scraper rings from 30 up to 200 mm nominal diameter Z-Rings	-	-	-
	IS 8422: Part 6: 1997 (Reaffirmed Year: 2019)	Piston rings for IC engines: Part 6 Slotted oil control rings from 50 up to 200 mm nominal diameter S-Rings	-	-	-
	IS 8422: Part 7: 1997 (Reaffirmed Year: 2019)	Piston rings for IC engines: Part 7 Double beveled slotted oil control rings from 50 up to 200 mm nominal diameter G-Rings	-	-	-
	IS 8422: Part 8: 1997 (Reaffirmed Year: 2019)	Piston rings for IC engines: Part 8 Narrow land slotted oil control rings from 50 up to 200 mm nominal diameter D-Rings	-	-	-
Finishing (Refer Table 4.a & 4.b)	IS 504: Parts 1 to 12: 2002 (Reaffirmed Year: 2018)	Methods of Chemical analysis of Aluminum and its alloys	USA	ASTM E3061 - 17	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Inductively Coupled Plasma Atomic Emission Spectrometry (Performance Based Method)
	IS 1608: Part 1: 2022	Metallic Materials — Tensile Testing Part 1 Method of Test at Room Temperature	USA	ASTM D7097 - 19	Standard Test Method for Determination of Moderately High Temperature Piston Deposits by Thermo-Oxidation Engine Oil Simulation Test—TEOST MHT
	IS 1608: Part 3: 2018	Metallic Materials — Tensile Testing Part 3 Method of Test at Low Temperature	USA	ASTM G206 - 17	Standard Guide for Measuring the Wear Volumes of Piston Ring Segments Run against Flat Coupons in Reciprocating Wear Tests

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
	-	-	USA	ASTM D6335 - 19	Standard Test Method for Determination of High Temperature Deposits by Thermo-Oxidation Engine Oil Simulation Tests

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	IS 5791: 2006 (Reaffirmed Year: 2020) - Internal Combustion Engines - Piston Rings - Material Specifications	- Table of typical materials - Hardness in terms of Vickers, Rockwell and Brinnell tests for the respective materials - Typical applications for respective categories of materials
	IS/ISO 6621: Part 1: 2018 - Internal Combustion Engines Piston Rings Part 1 Vocabulary (First Revision)	Glossary of terms
	IS/ISO 6621: Part 2: 2003 - internal Combustion Engines - Piston Rings Part 2 Inspection Measuring Principles	Terms most commonly used for piston rings
	IS 1387: 1993 (Reaffirmed Year: 2018) - General requirements for the supply of metallurgical materials	- General requirements for supply of metallurgical materials including raw as well as finished materials - General conditions for specific inspections and tests
	IS 1956: Part 1: 1976 - Glossary of terms relating to Iron and Steel – General metallurgy, Heat Treatment and testing	Glossary of terms related to General metallurgy, Heat Treatment and testing
	IS 5052: 1993 (Reaffirmed Year: 2022) - Aluminum and its alloys - temper designations	- Temper designations for cast and wrought products of aluminum and its alloys - Necessary Adjuncts
	IS 5047: Part 1: 1986 (Reaffirmed Year: 2022) - Glossary of terms relating to aluminum and aluminum alloys: Part 1 Unwrought and wrought metals	Glossary of terms (Necessary Adjunct to IS 5052: 1993 (Reaffirmed Year: 2022))
	IS 5047: Part 2: 1979 (Reaffirmed Year: 2022) - Glossary of terms relating to aluminum and aluminum alloys: Part 2 Plant and operations, thermal treatment, control and testing, finishing	Glossary of terms(Necessary Adjunct to IS 5052: 1993 (Reaffirmed Year: 2022))
	IS 10259: 1982 (Reaffirmed Year: 2020) - General conditions for delivery and inspection of aluminum and aluminum alloy products	General conditions of Delivery and inspection of aluminum and its alloys

Stage	Code	Scope
	ISO 18669 - 1: 2021 - Internal Combustion engines-Piston Pins, General Specifications	- Essential Dimension characteristics with a nominal outer diameter of 8mm to 100 mm - Vocabulary of terms
	ISO 6625: 1986 - Internal combustion engines — Piston rings — Oil control rings	- Vocabulary of terms - Measuring principles - Material Specifications
	IS 5791: 2006 (Reaffirmed Year: 2020) - Internal Combustion Engines - Piston Rings - Material Specifications	- Table of typical materials - Hardness in terms of Vickers, Rockwell and Brinnell tests for the respective materials - Typical applications for respective categories of materials
	IS/ISO 6621: Part 1: 2018 - Internal Combustion Engines Piston Rings Part 1 Vocabulary (First Revision)	Glossary of terms

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	IS 7793: 1975 (Reaffirmed Year: 2022) - Specification of Aluminum Alloys for I.C Engines	- Requirements of Aluminum Alloys for Internal Combustion Engine. - Sampling procedures for hyper eutectic alloys - Chemical analysis - Tensile and Hardness tests
	IS 11952: 1986 (Reaffirmed Year: 2019) - Specification for Steels for Piston Pins (Gudgeon Pins)	Chemical composition and mechanical properties
	IS IS 11998: 1987 (Reaffirmed Year: 2022) - Specification for Aluminum for the manufacture of alloys for I.C engine Pistons	- EC grade aluminum Ingots for manufacture of Eutectic and Hyper –Eutectic automotive pistons - Terminology
	IS 1820: 1979 (Reaffirmed Year: 2022) - Recommended shapes, Sizes and mass of aluminum notched bars for re-melting purposes	Specifies shapes, sizes and mass of notched bars and ingots used for remelting purposes in the industry

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	GSO ISO 2107: 2008 - Aluminum and Aluminum Alloys - Wrought Products - Temper Designations	Temper designations as required for identification for all product forms of wrought aluminum and aluminum alloys
	ISO 3547- 4: 2017 - Plain bearings — wrapped bushes — Part 4: Materials	- Specifications for solid and multi-layer bearing materials, such as those used for wrapped bushes in accordance with the other parts of ISO 3547 - Adjunct Standards

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	IS 734: 1975 (Reaffirmed Year: 2022) - Specification for wrought aluminum and aluminum alloy forging stock and forgings (For General Engineering purposes)	- Requirements for forging stock and forgings of aluminum and aluminum alloys in various conditions for general engineering - Chemical composition - Chemical and Mechanical tests - Tolerances
	IS 8422: Part 1: 1997 (Reaffirmed Year: 2019) - Plain compression rings from 30 up to 200 mm nominal diameter r - rings	- Specifies the dimensions, tolerances, tangential loads and other details of R-rings (plain: Compression rings) from 30 up to 200 mm nominal diameter for internal combustion engines - Dimensions and Tolerances
	IS 8422: Part 2: 1997 (Reaffirmed Year: 2019) - Taper Faced Compression Rings from 30 up to 200 mm Nominal Diameter M-rings	- Specifies dimensions, tolerances, tangential loads and other details of M-rings (taper faced: Compression rings) from 30 up to 200 mm nominal diameter for internal combustion engines - Dimensions and Tolerances
	IS 8422: Part 3: 1997 (Reaffirmed Year: 2019) - Keystone Rings from 82 up to 200 mm Nominal Diameter T- rings 15Å	- Specifies dimensions, tolerances, tangential loads and other details of M-rings (Keystone Rings) from 82 up to 200 mm nominal diameter T- Rings 15Å - Dimensions and Tolerances
	IS 8422: Part 4: 1997 (Reaffirmed Year: 2019) - Napier oil scraper rings from 30 up to 200 mm nominal diameter N-Rings	- Specifies the dimensions, tolerances, tangential loads and other details of N-rings (Napier oil scraper rings) from 30 up to 200 mm nominal diameter for internal combustion engines. - Dimensions and Tolerances
	IS 8422: Part 5: 1997 (Reaffirmed Year: 2019) - Stepped oil scraper rings from 30 up to 200 mm nominal diameter Z-Rings	- Specifies the dimensions, tolerances, tangential loads and other details of Z-rings (stepped oil scraper. Rings) from 30 up to 200 mm nominal diameter for internal combustion engines - Dimensions and Tolerances
	IS 8422: Part6: 1997 (Reaffirmed Year: 2019) - Piston rings for IC engines: Part 6 Slotted oil control rings from 50 up to 200 mm nominal diameter S-Rings	- Specifies the dimensions, tolerances, tangential loads and other details of S-rings (slotted oil control rings) from 50 up to 200 mm nominal diameter for internal combustion engines - Dimensions and Tolerances
	IS 8422: Part 7: 1997 (Reaffirmed Year: 2019) - Double beveled slotted oil control rings from 50 up to 200 mm nominal diameter G-Rings	- Specifies the dimensions, tolerances, tangential loads and other details of G-rings (double beveled slotted oil control rings) from 50 up to 200 mm nominal diameter for internal combustion engines - Dimensions and Tolerances
	IS 8422: Part 8: 1997 (Reaffirmed Year: 2019) - Piston rings for IC engines: Part 8 Narrow land slotted oil control rings from 50 up to 200 mm nominal diameter D-Rings	- Specifies the dimensions, tolerances, tangential loads and other details of D-rings (narrow land slotted oil control rings) from 50 up to 200 mm nominal diameter for internal combustion engines - Dimensions and Tolerances

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	ISO 3547-1: 2018 - Plain bearings — Wrapped bushes — Part 1: Dimensions	Specifies the dimensions and designations of cylindrical and flanged wrapped bushes made of mono and multi-layer bearing material for plain bearing applications
	ISO 16232: 2018 - Road vehicles — Cleanliness of components and systems	- Initial inspection and evaluation - Quality control or monitoring of manufacturing practices relevant to cleanliness
	ISO 18669 - 2: 2020 - Internal combustion engines — Piston pins — measuring principles	Measuring principles used for measuring piston pins
	ISO 4384-1: 2019 - Plain bearings — Hardness testing of bearing metals — Part 1: Multilayer bearings materials	- Parameters for the hardness testing of compound materials for plain bearings made from steel and bearing metal with bearing metals based on copper and aluminum, manufactured by casting, sintering or bonding - Adjunct standards and recommendations

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 504: Parts 1 to 12: 2002 (Reaffirmed Year: 2018) - Methods of Chemical analysis of Aluminum and its alloys	- Methods of determination of Silicon, Lead, Iron, Copper, Manganese, Magnesium, Nickel, Chromium, Tin, Antimony, Titanium and Zinc - Sampling and quality of Reagents - Test Methodology
	IS 11509: Part 2: 2012 (Reaffirmed Year: 2017) - Methods of tests for full flow lubricating oil filters for Internal combustion engines Part 1, Differential pressure/Flow characteristics	- Tests for determining the element by pass valve characteristics of full flow lubricating oil filters - Operational characteristics to be tested - Filter to be tested and diagrammatic representation of test rig

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	ASTM D7097 – 19 - Standard Test Method for Determination of Moderately High Temperature Piston Deposits by Thermo-Oxidation Engine Oil Simulation Test—TEOST MHT	- Prediction of the deposit-forming tendencies of engine oil in the piston ring belt and upper piston crown area. - Serves as a required test method to define API Category-Identified engine oils
	ASTM D6335 – 19 - Standard Test Method for Determination of High Temperature Deposits by Thermo-Oxidation Engine Oil Simulation Test	Predict the high temperature deposit forming tendencies of an engine oil subject to the added oxidizing stress of a turbocharger. This test method can be used to screen oil samples or as a quality assurance tool

Stage	Code	Scope
	ASTM E3061 – 17 - Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Inductively Coupled Plasma Atomic Emission Spectrometry (Performance Based Method)	Methods of inductively coupled plasma atomic emission spectrometric analysis of aluminum and aluminum alloys for Silicon, Manganese, Copper, Iron, magnesium, nickel etc.
	ISO 11881:1999 - Corrosion of metals and alloys — Exfoliation corrosion testing of aluminum alloys	Testing of wrought aluminum alloys in the form of semi-finished mill products and parts produced both from conventional ingot metallurgy processes and from powder metallurgy processes, as well as aluminum alloy metal matrix composites including those produced by mechanical alloying
	ASTM D6335 – 19 - Standard Test Method for Determination of High Temperature Deposits by Thermo-Oxidation Engine Oil Simulation Tests	- Profiling method for use accurately measuring the wear loss of compound-curved (crowned) piston ring specimens - Tests to evaluate the wear of materials, coatings, and surface treatments that are ideal for use in manufacture of piston rings and cylinder liners in diesel engines or spark ignition engines

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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3. Product information: https://en.wikipedia.org/wiki/Reciprocating_engine
4. Product information: https://en.wikipedia.org/wiki/History_of_the_internal_combustion_engine
5. Product information: <https://www.alcofilters.com/en-gb/products/oil-filters/>
6. Product information: <https://www.theengineerspost.com/types-of-piston-rings/>
7. Product information: https://en.wikipedia.org/wiki/Piston_rod
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9. Product information: <http://blog.wiseco.com/pistons-arent-round-profile-and-ovality-explained>
10. Product information: <http://blog.wiseco.com/what-is-forging>
11. Product information: <https://itstillruns.com/piston-manufacturing-process-5502005.html>
12. Product information: https://courses.washington.edu/engr100/Section_Wei/engine/UofWindsorManual/Piston%20and%20Piston%20Rings.htm
13. Domestic Legislation/Regulation: <http://www.kline.co.in/pdf/weight-restriction.pdf>, <https://www.certification-india.com/en/india-mandatory-bis-certification-for-vehicle-rims-planned/>
14. International Legislation/Regulation:
 - a. EU: https://ec.europa.eu/growth/sectors/automotive/legislation/motor-vehicles-trailers_en
<https://www.legislation.gov.uk/ukxi/2012/2619/contents/made>
 - b. USA: <https://www.law.cornell.edu/uscode/text/23/127>
<https://itstillruns.com/dot-specifications-rims-7423898.html>
 - c. AUS: [https://www.parliament.wa.gov.au/publications/tailedpapers.nsf/displaypaper/3912598a5d3063e216354de648257df100079558/\\$file/2598.pdf](https://www.parliament.wa.gov.au/publications/tailedpapers.nsf/displaypaper/3912598a5d3063e216354de648257df100079558/$file/2598.pdf)
https://www.infrastructure.gov.au/vehicles/vehicle_regulation/bulletin/vsb1/vsb_01_b.aspx
 - d. GULF (GCC): https://members.wto.org/crnattachments/2014/tbt/ARE/14_4732_00_e.pdf
15. Domestic Standards for Axel and Rims and raw materials: <https://standardsbis.bsbedge.com/>
16. ISO Standards: <https://www.iso.org/obp/ui#search>
17. International Standards:
 - a. EU: <https://www.en-standard.eu/>
 - b. USA: <https://www.en-standard.eu/astm-standards/> , <https://www.astm.org/>
 - c. AUS: <https://www.standards.org.au/>
 - d. GULF: <https://www.gso.org.sa/store/>
18. NABL accredited laboratories: <https://nabl-india.org/>, <https://parakh.ncog.gov.in>

HARROW & DISC HARROW, SHAMLI

1. INTRODUCTION

Agricultural tools and machinery are equipment's which are used to increase efficiency during farming operations. There are varieties of equipment's utilized for different functions ranging from tilling the soil, planting the seeds to harvesting the crops.



Shamli is known for the many MSME units involved in manufacturing of agricultural tools and implements. These units also export their products to various countries apart from supplying to the local demand.

The workshops produce a wide range of machinery and associated accessories. Clientele from surrounding states make up for a bulk of the market for these workshops. The agricultural implements being considered for the purpose of this compendium are:

- Disc Harrow
- Harrow

Harrow is an implement used to break up and smooth the surface of the soil.

It differs from the cultivator or tiller which are used for deeper tillage while the harrow is used to prepare the fields.

The difference between the Disc Harrow and Harrow being that the Harrow is chiefly used for lighter work such as levelling the aggregated clumps of soil, for this a harrow may comprise of a chain or springs while a Disc Harrow is used for heavy duty work such as breaking up the sod post ploughing.

2. RAW MATERIALS

- Harrows are rugged farming implements and they can be made of different materials depending on the soil conditions. The raw materials used can vary from conventional steels to high carbon steel. Polish and paints are used for providing specific colours.
- Chain Link mesh
- Chains
- Discs made of wrought Steel, Carbon Steel etc. with edges curved and sharpened
- Axles

3. MANUFACTURING

The basic steps in the manufacturing of a Harrow are:

3.1. Harrow

- A simple harrow is fabricated from iron parts. The basic

process commences with the fabrication of a frame.

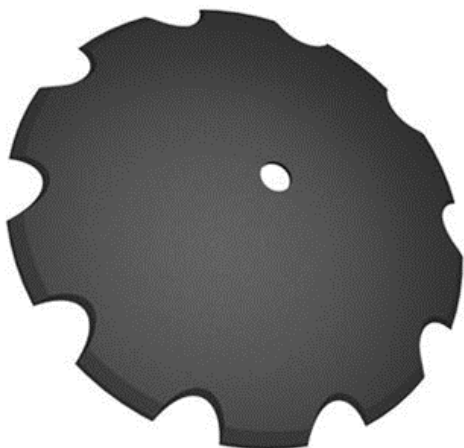


Figure 1: Harrow Disc

- Depending on the use whether for agricultural or horticultural use, the frame may be made robust or out of lighter materials. The metal sections for the frame are cut to the desired size and welded together to form a box.



Figure 2: Chain Mesh Harrow

- Next the rigid assembly is fabricated and holes are drilled into it for placing bolts for attaching it to the tractor. For lighter horticultural use, chains may be attached to the frame

which can then be attached to the tractor

- A heavy duty chain link mesh is welded to the frame
- The fabricated welds are sanded properly and a final coat of paint is applied on the finished harrow.

3.2. Disc Harrow

- A Disc Harrow is fabricated from iron parts. The basic process commences with the preparation of the frame.



Figure 3: Disc Harrow

- Rectangular hollow pipe sections are selected as per the length desired and are cut. These are then welded to form a box.
- Next the rigid assembly is fabricated and holes are drilled into it for placing bolts for attaching it to the tractor.
- Steel channel sections are cut and bent to form the mounting supports for the axles carrying the circular discs.

- These are placed at angles away from each other to cover a wider area.
- The specially manufactured circular discs are now mounted on the axles using spacers with bearings (Which are used to attach the axles to the frame) and the either ends of the axle are secured using lock nuts.

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check for the dimensions
- Check the welded joints for proper welding
- The cultivator shoes should be attached firmly to the frame
- The fabricated components should not have any foreign substances adhering to the surface
- Paint should be applied uniformly and should not be flaking off

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the

overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organizations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce

(<https://commerce.gov.in/international-trade/trade-agreements/>).

5.1. Domestic

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	India	NA	NA	NA

5.2. International

S. No.	Region/Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Safeguard Decision 2013/426 (as amended)	a. European Commission	a. Phytosanitary certification b. Inspection for Animal Diseases
2	United States of America	a. Code of Federal Regulations 9CFR94.1 b. APHIS Regulations	a. APHIS (Animal & Plant Health Inspection Service)	a. Inspection of Animal Diseases b. Phytosanitary requirements
3	Australia	a. Cleanliness requirements b. Bio Security Import Condition System (BICON)	a. Department of Agriculture, Water and Environment	a. Machinery not meeting the acceptable cleanliness standards of the Department of Agriculture and Water Resources may not be permitted entry into Australia b. Biosecurity import conditions

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
General (Refer Table 1)	IS 9818: Part 2: 1981	Glossary of terms related to tillage and inter-cultivation equipment	Australia	AS/NZS 2153.1:1997. ISO 4254-1:1989	Tractors and machinery for agriculture and forestry— Technical means for ensuring safety
	IS 620: 1985 (Reaffirmed Year: 2020)	Specification for (reaffirmed 2010) wooden tool handles general requirements	European Union	BS BS EN 709 - 1997+A4 - 2009	Safety requirements, test methods, methods for elimination or reduction of risks arising from their use
	IS 707: 2011 (Reaffirmed Year: 2021)	Timber technology and utilization of wood, bamboo and cane - Glossary of terms (Adjunct standard to IS 620: 1985 (Reaffirmed Year: 2020))		ISO 4254-5: 2018	Agricultural machinery — Safety — Part 5: Power-driven soil-working machines
	IS 4970: 1973 (Reaffirmed Year: 2020)	Key to identification of commercial timber, (Adjunct to IS 620: 1985 (Reaffirmed Year: 2020))	-	-	-
	IS 6635: 1972 (Reaffirmed Year: 2021)	Tractor operated Disc Harrow	-	-	-

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
Raw Materials (Refer Table 2)	IS 7201: Part 1: 1987 (Reaffirmed Year: 2019)	Methods of sampling for agricultural machinery and equipment (Hand drawn/animal drawn equipment)	-	-	-
	IS 7230: 1974 (Reaffirmed Year: 2021)	Plain Spool for tractor operated Disc Harrow	-	-	-
	IS 10225: 1982 (Reaffirmed Year: 2018)	Bearing Spools for Tractor-operated Disc Harrows	-	-	-
	IS 2609: 1972 (Reaffirmed Year: 2019)	Specification for coach bolts (Material specification for attaching bolts, Table II)	-	-	-
	IS 1570: Part 1: 1978 (Reaffirmed Year: 2019)	Schedules of Wrought Steel for General Engineering Purposes	-	-	-
Manufacturing/ Processing (Refer Table 3.a & 3.b)	IS 822: 1970 (Reaffirmed Year: 2019)	Code of procedure for inspection of welds		ISO 730: 2009	Agricultural wheeled tractors — Rear-mounted three-point linkage — Categories 1N, 1, 2N, 2, 3N, 3, 4N and 4
	IS 4366: Part 1: 1985 (Reaffirmed Year: 2021)	Specification for Agricultural tillage discs, Part 1: Concave types	-	-	-
	IS 4468: Part 1: 1997 (Reaffirmed Year: 2022)	Dimensions for three-point linkage of agricultural wheeled tractors	-	-	-
Finishing (Refer Table 4)	IS 1500: Part 2: 2021	Method for Brinell test for metallic materials	-	-	-
	IS 7640: 1975 (Reaffirmed Year: 2019)	Test Code for Disc Harrows	-	-	-
	IS 9217: 1979 (Reaffirmed Year: 2021)	Test Code for Agricultural Discs	-	-	-

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	AS/NZS 2153.1 – 1997 - Tractors and machinery for agriculture and forestry— Technical means for ensuring safety	- This standard provides guidelines regarding the prevention of accidents arising from the use of tractors and machinery for agriculture and forestry - It also specifies technical means of improving the degree of personal safety of operators and others involved in the course of normal running, maintenance and use, intended to be carried out by the user of the machinery - This part of the standard gives general guidelines to be met when designing tractors and machinery for agriculture and forestry
	IS 9818: Part 2: 1981 - Glossary of terms related to tillage and inter-cultivation equipment	Glossary of terms
	IS 620: 1985 (Reaffirmed Year: 2020) - Specification for (reaffirmed 2010) wooden tool handles general requirements	- Specification of types of woods to be used - General requirements of different type of tool handles - Classification of tool handles
	IS 707: 2011 (Reaffirmed Year: 2021) - Timber Technology and Utilization of Wood, Bamboo and Cane - Glossary of Terms	Adjunct to IS 620: 1985 (Reaffirmed Year: 2020), Timber Technology and Utilization of Wood, Bamboo and Cane - Glossary of Terms
	IS 4970: 1973 (Reaffirmed Year: 2020) - Key to identification of Commercial timber	Key to identification of Commercial timber, Adjunct to IS 620: 1985 (Reaffirmed Year: 2020)
	IS 6635: 1972 (Reaffirmed Year: 2021) - Tractor operated Disc Harrow	- Prescribes Material and specifications for Tractor operated Disc Harrow - Adjunct Standards
	IS 4905: 2015 (Reaffirmed Year: 2020) - Methods of Random Sampling	Method of selection of tractor-mounted spring-loaded cultivator
	IS 7201: Part 1: 1987 (Reaffirmed Year: 2019) - Methods of sampling for Agricultural machinery and equipment	Methods of sampling for agricultural machinery and equipment (Hand drawn/Animal Drawn equipment)
	BS EN 709 - 1997+A4 – 2009 - Safety requirements, test methods, methods for elimination or reduction of risks arising from their use	Safety requirements, test methods, methods for elimination or reduction of risks arising from their use
	ISO 4254-5: 2018 - Agricultural machinery — Safety — Part 5: Power-driven soil-working machines	Basic Principles of Safety

6.2. Table 2: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	IS 7230: 1974 (Reaffirmed Year: 2021) - Plain Spool for tractor operated Disc Harrow	Specifies materials, dimensions and other requirements for Plain Spool for tractor operated Disc Harrow
	IS 10225: 1982 (Reaffirmed Year: 2018) - Bearing Spools for Tractor-operated Disc Harrows	- Specifies materials, dimensions and other requirements for Bearing Spool for tractor operated Disc Harrow - Adjunct Standards
	IS 1570: Part 1: 1978 (Reaffirmed Year: 2019) - Schedules of Wrought Steel for General Engineering Purposes	- Schedules of Wrought Steel - Adjunct Standards
	IS 2609: 1972 (Reaffirmed Year: 2019) - Specification for coach bolts	Requirements for coach bolts in the size range 6 to 24 mm

6.3. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	IS 822: 1970 (Reaffirmed Year: 2019) - Code of procedure for inspection of welds	- Recommended procedures for Inspection of Welds - Exclusions
	IS 4366: Part 1: 1985 (Reaffirmed Year: 2021) - Specification for Agricultural tillage discs, Part 1: Concave types	Specifies the materials, dimensions and other requirements for agricultural tillage discs, concave type
	IS 4468: Part 1: 1997 (Reaffirmed Year: 2022) - Dimensions for three-point linkage of agricultural wheeled tractors	- Specifies the dimensions and requirements of the three-point linkage for the attachment of implements of equipment to the rear of agricultural wheeled tractors - Adjunct Standards

6.4. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	ISO 730: 2009 - Agricultural wheeled tractors — Rear-mounted three-point linkage — Categories 1N, 1, 2N, 2, 3N, 3, 4N and 4	- Dimensions and Requirements for Three Point Linkages - Normative References - Definitions

6.5. Table 4: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 1500: Part 2: 2021 - Brinell test for testing hardness of shovel and sweep	Brinell test for hardness of shovel and sweep
	IS 7640: 1975 (Reaffirmed Year: 2019) - Test Code for Disc Harrows	- Test Methods for evaluation of Disc Harrows - Adjunct Standards
	IS 9217: 1979 (Reaffirmed Year: 2021) - Test Code for Agricultural Discs	- Test Code for Agricultural Discs - Adjunct Standards

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

REFERENCES

1. Product information: <http://odopup.in/en/article/ghaziabad>
2. Product information: <http://www.harrowdiscmanufactures.com/notched-disc-blades.php>
3. Product information: <https://www.countrysmallholding.com/news/five-reasons-to-harrow-6258796>
4. Product information: <https://toptillers.com/harrowing-guide/>
5. International Legislation/Regulation:
AUS: <https://bicon.agriculture.gov.au/BiconWeb4.0>
6. Domestic Standards for Harrow and raw materials: <https://standardsbis.bsbedge.com/>
7. BIS product conformation overview: <https://bis.gov.in>
8. ISO Standards: <https://www.iso.org/obp>
9. International Standards:
 - a. EU: <https://www.en-standard.eu/>
 - b. USA: <https://www.en-standard.eu/astm-standards/>, <https://www.astm.org/>
 - c. AUS: <https://www.standards.org.au/>
 - d. GULF: <https://www.gso.org.sa/store/>
10. NABL accredited laboratories: <https://nabl-india.org/>, <https://parakh.ncog.gov.in>

APPENDIX I: TESTING AND CERTIFICATION

Uttar Pradesh is home to a large number of indigenous products and manufacturing units associated with these unique products. The crafts and artisans have both withstood the vagaries of time and have flourished under the patronage provided in the form of various schemes to promote these products and bring about innovation in the industry by up skilling the artisans as well as providing them assistance in tapping new markets and becoming competitive in the modern market by utilizing machinery developed locally to augment their efforts.

With the advent of machinery and technology there has been a gradual move towards standardization to improve both the product life cycle as well as the quality of materials used. Two important aspects in standardization of products and processes are:

- Testing (Product/Material testing)
- Certification (Product Certification or Systems Certification)

1. PRODUCT TESTING

What is product testing?

Testing is the determination of one or more of a product's characteristics and is usually performed by a laboratory. In other words, product testing is a process to measure its performance, safety, quality, and compliance with established standards.

Quality testing is involved at each step of the production process and often

begins with the testing of raw materials, moving through the testing of in-process products to the finished product.

Testing is an essential part of developing a quality product. Any product that is produced is expected to meet certain requirements and specifications to ensure standardization and consistency.

Why is testing of products required?

Testing is an essential part of developing a quality product. Product testing is required to assure quality, performance and safety of the product. Product testing is also required to comply with the regulatory requirements and standards, wherever prescribed.

Testing at the various stages of production helps to ensure that the products meet the defined specifications and gain acceptance globally.

Why should you get the product testing done from an approved/ accredited laboratory?

It is always recommended to get the tests done at a Laboratory that is approved /accredited by a competent authority. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the

International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Testing and Calibration Laboratories (NABL), a constituent Board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such laboratory. NABL is Mutual Recognition Arrangements (MRA) signatory to ILAC as well as APAC for the accreditation of Testing and Calibration Laboratories (ISO/IEC 17025), Medical Testing Laboratories (ISO 15189), Proficiency Testing Providers (PTP) (ISO/IEC 17043) and Reference materials producers (RMP).

Such MRA reduces technical barrier to trade and facilitates acceptance of test/calibration results between countries which MRA partners represent.

Further, for certain products like food, pharmaceuticals etc., there are regulatory requirements (domestic as well as region or country specific (if exporting)) for testing of

the raw materials and finished goods which are mandatory to be followed. For such products, various regulations and the standards making bodies like Bureau of Indian Standards (BIS) also notify specific Laboratories, from time to time. For e.g. BIS has recognised several laboratories for the tests prescribed under the Indian Standards formulated by them. Similarly, other regions or countries may have specific regulations that require prescribed tests, for e.g. REACH regulation of the European Union, specifies requirements to for the protection of human health and the environment from the risks that can be posed by chemicals.

Where can you get the list of accredited/approved laboratories?

List of the testing laboratories for the relevant tests may be obtained from the following URLs:

- **National Accreditation Board for Testing and Calibration Laboratories (NABL):** <https://nabl-India.org/>
- **Bureau of Indian Standards (BIS):** <https://bis.gov.in/>

2. CERTIFICATION

Certification is a provision by an independent body of written assurance (a certificate) that the product, service or system in question meets specific requirements. Broadly, certifications may be categorised as:

- a. **Management Systems Certification:** These certifications help organizations improve their performance by specifying

repeatable steps that organizations consciously implement to achieve their goals and objectives, and to create an organizational culture that reflexively engages in a continuous cycle of self-evaluation, correction and improvement of operations and processes through heightened employee awareness and management leadership and commitment.

The benefits of an effective management system to an organization include:

- More efficient use of resources and improved financial performance
- Improved risk management and protection of people and the environment
- Increased capability to deliver consistent and improved services and products, thereby increasing value to customers and all other stakeholders

Examples of Systems Certification include ISO 9001 QMS, ISO 22000 FSMS, etc.

b. Product Certification: Product certification is the process of certifying that a specific product fulfils requirements prescribed in contracts, regulations or specifications and has passed all necessary quality assurance and performance tests. Product certification is often required in industries and marketplaces where

product failure could bring serious negative consequences to the health and safety of people and environment. The benefits of an effective management system to an organization include:

- Product certification gives product users the confidence they need to purchase the desired products.
- If a product is certified according to applicable national or international laws, customers know that that product will function correctly and won't cause any harm. Examples of product certification include CE marking, ISI mark, Hallmark, etc.

Are all certifications mandatory?

Mostly, the certifications are voluntary in nature. However, for a number of products, compliance to specific Standards may be made compulsory, for instance, compliance to Indian Standards for a number of products, has been made compulsory by the Central Government under various considerations viz. public interest, protection of human, animal or plant health, safety of environment, prevention of unfair trade practices and national security. For such products, the Central Government directs mandatory use of Standard Mark under a Licence or Certificate of Conformity (CoC) from BIS through issuance of Quality Control Orders (QCOs). Similarly, other regions or countries may have specific regulations that

require prescribed certifications, for e.g. USFDA prescribes specific requirements for Food, Drugs and Cosmetic products.

Why should you get the Certification from an accredited Certification Body?

It is always recommended to obtain the certification from accredited Certification Bodies or the ones that are prescribed by the relevant regulatory authorities. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Certification Bodies (NABCB), a constituent board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such Certification and Inspection Bodies. The National Accreditation Board for Certification Bodies provides accreditation to Certification and Inspection Bodies based on assessment of their competence as per the Board's criteria and in accordance with International Standards and Guidelines.

NABCB is internationally recognized and represents the interests of the Indian industry at international forums through membership and active participation with the objective of becoming a signatory to international Multilateral / Mutual Recognition Arrangements (MLA / MRA).

Where can you get the list of accredited/approved laboratories?

List of the NABCB accredited Certification Bodies (CBs) and Inspection Bodies (IBs) for relevant certifications may be obtained from the following URL:

National Accreditation Board for Certification Bodies (NABCB):
<http://nabcb.qci.org.in/>

APPENDIX II: IMPORTANT INSTITUTIONS

ENGINEERING GOODS				
S. No.	Institute	Areas of Expertise	Postal Address	Contact
1	Indian Institute of Technology- Banaras Hindu University - Department of Ceramic Engineering - Department of Metallurgical Engineering - Department of Mechanical Engineering	- Machine Design - Thermal Fluid engineering - Production Engineering - Industrial Management - Corrosion - Metal Fatigue, wear, friction - Advanced Metals - Physical Metallurgy of Alloys - Glass & Glass Ceramics - Electrical Ceramics - Refractories	IIT- BHU, Ajagara, Varanasi, Uttar Pradesh 221005	Email: head.cer@iitbhu.ac.in Phone: 9936182124 Website: www.iitbhu.ac.in/dept/cer Email: head.met@iitbhu.ac.in Phone: 05427081-228 Website: www.iitbhu.ac.in/dept/met Email: head.mec@iitbhu.ac.in Phone: 0542-2368157 Website: www.iitbhu.ac.in/dept/mec
2	Indian Institute of Technology - Kanpur - Department of Materials Science & Engineering - Department of Mechanical Engineering - Department of Industrial & Management Engineering	- Manufacturing processes - Mechanical Behavior of Metals - Physical Metallurgy - Extractive Metallurgy - Structural Ceramics - Technology Development - Advanced manufacturing processes - Management Engineering focused on Improvement of Quality, Productivity and Profitability - Consulting Projects	IIT Kanpur, Kalyanpur, Kanpur, Uttar Pradesh - 208016	Email: gkumar@iitk.ac.in Phone: 0512 259 7640 Website: www.iitk.ac.in/mse Email: head.met@iitk.ac.in Phone: 05427081-228 Website: www.iitk.ac.in/me Email: head_ime@iitk.ac.in Phone: 0512- 6796606 Website: www.iitk.ac.in/ime
3	International Advanced Research Centre for Powder Metallurgy and New Materials (ACRI)	- Nano materials - Ceramics - Surface Engineering - Laser Processing	International Advanced Research Centre for Powder Metallurgy and New Materials (ACRI), Balapur Post Office, Hyderabad - 500005, Telangana, India	Email: info@arci.res.in Phone: +91 - 40 - 2445 2200 Website: https://arci.res.in/

